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Part I
preliminaries

1. Basics of Magnetism

1.1 Direct Exchange

1.1.1 Coulomb exchange

We are looking for a mechanism that can explain magnetic ordering at high temperatures. One might expect that this requires a spin-dependent interaction, however, no such interaction of this strength is known. This is where direct exchange enters the stage. It is made up of multiple mechanics, the first one (canonically) being Coulomb exchange. This idea goes back to Heisenberg. It is able to explain how ferromagnetic order comes about in a system of localized spins (e.g. spins of unpaired valence electrons). More precise? The most remarkable thing: the story starts with the (spin-indepedent!) Coulomb interaction between the electrons. In second quantization the Coulomb interaction reads

$$H_C = \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0}\int d^3r_1 \int d^3r_2 \sum_{\sigma_1,\sigma_2} \Psi^\dagger_{\sigma_1}(\vec{r}_1)\Psi^\dagger_{\sigma_2}(\vec{r}_2)\frac{1}{|\vec{r}_1-\vec{r}_2|}\Psi_{\sigma_2}(\vec{r}_2)\Psi_{\sigma_1}(\vec{r}_1). \tag{1}$$

$\Psi^\dagger_{\sigma_1}(\vec{r}_1)$ is a field operator that creates an electron with spin σ_1 at position $\vec{r}_1$. These are now expanded in orthonormal wavefunctions localized around the ions (Wannier functions) $\phi_{\vec{R},m}(\vec{r})$ and spinors χ_σ:

$$\Psi^\dagger_{\sigma_1}(\vec{r}_1) = \sum_{\vec{R},m} a^\dagger_{\vec{R}m\sigma_1}\phi_{\vec{R}m}(\vec{r}_1)\chi_{\sigma_1}. \tag{2}$$

The sum over $\vec{R}$ goes over the positions of all Ions. m counts all angular degrees of freedom, but not the spin. $a^\dagger_{\vec{R},m,\sigma_1}$ is a fermionic creation operator. One finds

$$\begin{aligned} H_C = \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0} \sum_{\substack{\vec{R}^{(1)}\dots\vec{R}^{(4)}\\ m^{(1)}\dots m^{(4)}}} \int d^3r_1 \int d^3r_2\, \phi^*_{\vec{R}^{(1)}m^{(1)}}(\vec{r}_1)\phi^*_{\vec{R}^{(2)}m^{(2)}}(\vec{r}_2)\frac{1}{|\vec{r}_1-\vec{r}_2|}\phi_{\vec{R}^{(3)}m^{(3)}}(\vec{r}_2)\phi_{\vec{R}^{(4)}m^{(4)}}(\vec{r}_1) \\ \times \sum_{\sigma_1,\sigma_2} \chi^\dagger_{\sigma_1}\chi^\dagger_{\sigma_2}\chi_{\sigma_2}\chi_{\sigma_1} a^\dagger_{\vec{R}^{(1)}m^{(1)}\sigma_1} a^\dagger_{\vec{R}^{(2)}m^{(2)}\sigma_2} a_{\vec{R}^{(3)}m^{(3)}\sigma_2} a_{\vec{R}^{(4)}m^{(4)}\sigma_1}. \end{aligned} \tag{3}$$

The expression is simplified by evaluating the spinor product.

$$\begin{aligned} H_C = \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0} \sum_{\substack{\vec{R}^{(1)}\dots\vec{R}^{(4)}\\ m^{(1)}\dots m^{(4)}\\ \sigma_1,\sigma_2}} \left\langle \vec{R}^{(1)}m^{(1)}, \vec{R}^{(2)}m^{(2)} \left| \frac{1}{|\vec{r}_1-\vec{r}_2|} \right| \vec{R}^{(3)}m^{(3)}, \vec{R}^{(4)}m^{(4)} \right\rangle \\ \times a^\dagger_{\vec{R}^{(1)}m^{(1)}\sigma_1} a^\dagger_{\vec{R}^{(2)}m^{(2)}\sigma_2} a_{\vec{R}^{(3)}m^{(3)}\sigma_2} a_{\vec{R}^{(4)}m^{(4)}\sigma_1} \end{aligned} \tag{4}$$

It is worthwile to take a few detours in order to explore different effects of the Coulomb interaction, increasing the complexity in the mean-time.

on-site, single orbital: Hubbard U-term

First, focus on the on-site effects $\vec{R}^{(1)} = \vec{R}^{(2)} = \vec{R}^{(3)} = \vec{R}^{(4)}$, dropping the position quantum numbers. The easiest case is that of a single orbital per site, $m^{(1)} = m^{(2)} = m^{(3)} = m^{(4)} \equiv m$.

$$\Rightarrow H_C = \frac{1}{2}\sum_{\vec{R}}\sum_{\sigma_1\sigma_2} \left\langle \frac{1}{|\vec{r}_1-\vec{r}_2|} \right\rangle_{m,m} a^\dagger_{\sigma_1} a^\dagger_{\sigma_2} a_{\sigma_2} a_{\sigma_1} \tag{5}$$

The position indices of the ladder operators have been supressed.

$$\sum_{\sigma_1\sigma_2} a^\dagger_{\sigma_1} a^\dagger_{\sigma_2} a_{\sigma_2} a_{\sigma_1} = a^\dagger_\downarrow a^\dagger_\uparrow a_\uparrow a_\downarrow + a^\dagger_\uparrow a^\dagger_\downarrow a_\downarrow a_\uparrow = 2a^\dagger_\downarrow a^\dagger_\uparrow a_\uparrow a_\downarrow \tag{6}$$

Two out of the four terms ($\sigma_1 = \sigma_2 = \uparrow$ and $\sigma_1 = \sigma_2 = \downarrow$) are identically zero, since $(a_{\downarrow/\uparrow})^2 = 0$. In other words: due to Pauli principle the spin allignement is already fixed. So no exchange interaction can arise here. The technical reason is that we assumed a single orbital on a single site (2 states), leaving the electrons no other choice. We'll have to loosen the approximations to see exchange play out. Let's close with one last observation regarding the one-site, single orbital story;

$$\Rightarrow H_{\mathrm{C}} = \sum_{\vec{R}} U a^\dagger_\downarrow a^\dagger_\uparrow a_\uparrow a_\downarrow, \quad U = \frac{e^2}{4\pi\varepsilon_0} \int d^3r_1 \int d^3r_2 \, |\phi(\vec{r}_1)|^2 \frac{1}{|\vec{r}_1 - \vec{r}_2|} |\phi(\vec{r}_2)|^2 \tag{7}$$

This gives the Hubbard U-term. It's interpretation is pretty straight forward: electrons occupying the same orbital have increased Coulomb energy, given by U. This presents a convenient way to include Coulomb interaction in tight-binding models. It is only an approximation, since all other terms (multiple sites, different orbitals) have been ignored so far. It is also clear that this contribution is the largest off all effects arising from the Coulomb interaction as the overlap is biggest here.

on-site, multiple orbitals: Hund's rule

The first step of complexification is to allow multiple orbitals at every ion, thereby reintroducing the $m^{(i)}$ quantum numbers. If one treats the Coulomb interaction perturbatively, in first order the creation/ annihilation operators have to be grouped into pairs in order to produce non-vanishing contributions. I am not satisfied with this. There are two possibilities: $m^{(1)} = m^{(3)}$, $m^{(2)} = m^{(4)}$ and $m^{(1)} = m^{(4)}$, $m^{(2)} = m^{(3)}$.

$$\begin{aligned} K_{m^{(1)}m^{(2)}} &:= \frac{e^2}{4\pi\varepsilon_0} \left\langle m^{(1)}, m^{(2)} \left| \frac{1}{|\vec{r}_1 - \vec{r}_2|} \right| m^{(2)}, m^{(1)} \right\rangle \\ &= \frac{e^2}{4\pi\varepsilon_0} \int d^3r_1 \int d^3r_2 \, |\phi_{m^{(1)}}(\vec{r}_1)|^2 \frac{1}{|\vec{r}_1 - \vec{r}_2|} |\phi_{m^{(2)}}(\vec{r}_2)|^2 \end{aligned} \tag{8}$$

$$\begin{aligned} J_{m^{(1)}m^{(2)}} &:= \frac{e^2}{4\pi\varepsilon_0} \left\langle m^{(1)}, m^{(2)} \left| \frac{1}{|\vec{r}_1 - \vec{r}_2|} \right| m^{(1)}, m^{(2)} \right\rangle \\ &= \frac{e^2}{4\pi\varepsilon_0} \int d^3r_1 \int d^3r_2 \, \phi^*_{m^{(1)}}(\vec{r}_1) \phi^*_{m^{(2)}}(\vec{r}_2) \frac{1}{|\vec{r}_1 - \vec{r}_2|} \phi_{m^{(1)}}(\vec{r}_2) \phi_{m^{(2)}}(\vec{r}_1) \end{aligned} \tag{9}$$

$$\begin{aligned} \Rightarrow H_C \approx \frac{1}{2} \sum_{\vec{R}} \sum_{\substack{m^{(1)},m^{(2)} \\ \sigma_1,\sigma_2}} \Big(& K_{m^{(1)}m^{(2)}} a^\dagger_{m^{(1)}\sigma_1} a^\dagger_{m^{(2)}\sigma_2} a_{m^{(2)}\sigma_2} a_{m^{(1)}\sigma_1} \\ & + J_{m^{(1)}m^{(2)}} a^\dagger_{m^{(1)}\sigma_1} a^\dagger_{m^{(2)}\sigma_2} a_{m^{(1)}\sigma_2} a_{m^{(2)}\sigma_1} \Big) \end{aligned} \tag{10}$$

The position indices have been supressed! The exchange term may be reexpressed as

$$\begin{aligned} -\sum_{\vec{R}} \sum_{m^{(1)},m^{(2)}} J_{m^{(1)}m^{(2)}} \Big(& \left(a^\dagger_{m^{(1)}\uparrow} a_{m^{(1)}\uparrow} - a^\dagger_{m^{(1)}\downarrow} a_{m^{(1)}\downarrow} \right) \left(a^\dagger_{m^{(2)}\uparrow} a_{m^{(2)}\uparrow} - a^\dagger_{m^{(2)}\downarrow} a_{m^{(2)}\downarrow} \right) \\ & + a^\dagger_{m^{(1)}\uparrow} a_{m^{(1)}\downarrow} a^\dagger_{m^{(2)}\downarrow} a_{m^{(2)}\uparrow} + a^\dagger_{m^{(1)}\downarrow} a_{m^{(1)}\uparrow} a^\dagger_{m^{(2)}\uparrow} a_{m^{(2)}\downarrow} \\ & + a^\dagger_{m^{(1)}\uparrow} a_{m^{(1)}\uparrow} a^\dagger_{m^{(2)}\downarrow} a_{m^{(2)}\downarrow} + a^\dagger_{m^{(1)}\downarrow} a_{m^{(1)}\downarrow} a^\dagger_{m^{(2)}\uparrow} a_{m^{(2)}\uparrow} \Big). \end{aligned} \tag{11}$$

Technically there are terms missing from the Fermi-algebra. However, these are only bilinear and thus act as a siengle-particle potential and are not relevant here. With the mapping/ substitution

$$n_m = a^\dagger_{m\uparrow}a_{m\uparrow} + a^\dagger_{m\downarrow}a_{m\downarrow}, \quad s_{mz} = \frac{1}{2}\left(a^\dagger_{m\uparrow}a_{m\uparrow} - a^\dagger_{m\downarrow}a_{m\downarrow}\right), \quad s_{m+} = a^\dagger_{m\uparrow}a_{m\downarrow} \tag{12}$$

one ends up with

$$H_\mathrm{C} = \frac{1}{2}\sum_{\vec{R}}\sum_{m^{(1)},m^{(2)}}\left(K_{m^{(1)}m^{(2)}} - \frac{1}{2}J_{m^{(1)}m^{(2)}}\right)n_{m^{(1)}}n_{m^{(2)}} - 2J_{m^{(1)}m^{(2)}}\vec{s}_{m^{(1)}}\cdot\vec{s}_{m^{(2)}}. \tag{13}$$

The first term is the on-site Coulomb interaction between electrons in different orbitals on the same site. Additionally, $K_{m^{(1)}m^{(2)}} - \frac{1}{2}J_{m^{(1)}m^{(2)}} > 0$, meaning the Coulomb interaction remains repulsive. Returnung to the exchange term; using

$$\frac{1}{|\vec{r}_1 - \vec{r}_2|} = \int \frac{d^3k}{(2\pi)^3}\frac{4\pi}{k^2}e^{i\vec{k}(\vec{r}_1-\vec{r}_2)} \tag{14}$$

$$\Rightarrow J_{m^{(1)}m^{(2)}} = \frac{e^2}{\varepsilon_0}\int\frac{d^3k}{(2\pi)^3}\frac{1}{k^2}\underbrace{\int d^3r_1\ \phi^*_{m^{(1)}}(\vec{r}_1)\phi_{m^{(2)}}(\vec{r}_1)e^{i\vec{k}\vec{r}_1}}_{I(\vec{r})}\underbrace{\int d^3r_2\ \phi_{m^{(1)}}(\vec{r}_2)\phi^*_{m^{(2)}}(\vec{r}_2)e^{-i\vec{k}\vec{r}_2}}_{I^*(\vec{r})}. \tag{15}$$

Hence, $J_{m^{(1)}m^{(2)}}$ is positive. Therefore the interaction is minimized by ferromagnetic allignement. This is the derivation of Hund's first rule: the ground state of a partially filled shell is the one that maximizes $\vec{s}_\mathrm{tot}$.

different site, single orbital: Coulomb exchange

Assuming again singular orbitals $m^{(1)} = m^{(2)} = m^{(3)} = m^{(4)} \equiv m$, whilst reinstating the position quantum numbers the calculation is symmetrical to that in the last subchapter.

$$\Rightarrow H_\mathrm{C} = \frac{1}{2}\sum_{\vec{R}^{(1)},\vec{R}^{(2)}}\left(K_{\vec{R}^{(1)}\vec{R}^{(2)}} - \frac{1}{2}J_{\vec{R}^{(1)}\vec{R}^{(2)}}\right)n_{\vec{R}^{(1)}}n_{\vec{R}^{(2)}} - 2J_{\vec{R}^{(1)}\vec{R}^{(2)}}\vec{s}_{\vec{R}^{(1)}}\cdot\vec{s}_{\vec{R}^{(2)}} \tag{16}$$

$$K_{\vec{R}^{(1)}\vec{R}^{(2)}} = \frac{e^2}{4\pi\varepsilon_0}\int d^3r_1\int d^3r_2\left|\phi_{\vec{R}^{(1)}}(\vec{r}_1)\right|^2\frac{1}{|\vec{r}_1-\vec{r}_2|}\left|\phi_{\vec{R}^{(2)}}(\vec{r}_2)\right|^2 \tag{17}$$

$$J_{\vec{R}^{(1)}\vec{R}^{(2)}} = \frac{e^2}{4\pi\varepsilon_0}\int d^3r_1\int d^3r_2\phi^*_{\vec{R}^{(1)}}(\vec{r}_1)\phi^*_{\vec{R}^{(2)}}(\vec{r}_2)\frac{1}{|\vec{r}_1-\vec{r}_2|}\phi_{\vec{R}^{(1)}}(\vec{r}_2)\phi_{\vec{R}^{(2)}}(\vec{r}_1) \tag{18}$$

As in the last chapter $J_{\vec{R}^{(1)}\vec{R}^{(2)}} > 0$, so again the interaction is ferromagnetic. The interpretation is the same as for the previous subchapter; electrons in a triplet state avoid each other in real space due to Pauli-principle ("exchange hole"), thereby reducing their Coulomb energy.

In terms of explaining magnetic ordering this obviously can't be the end of the story, since we haven't found a way to explain anti-ferromagnetic order yet.

2. Partons

2.1 SU(2) Slave Fermions

The method described here to represent spins in terms of Fermions was first employed by Abrikosov in 1965 and later reformulated in terms of gauge theory by Affleck in 1988. The idea (for spin 1/2!) is to introduce two new flavours of fermions: $c^\dagger_{i\uparrow}$, $c^\dagger_{i\downarrow}$ on each site. The transformation reads

$$\vec{S}_i = \frac{1}{2} \sum_{\alpha,\beta} c^\dagger_{i\alpha} \vec{\sigma}_{\alpha\beta} c_{i\beta}, \tag{19}$$

where $\vec{\sigma}$ is the vector of Pauli matrices.

This transformation is quite natural, as it has been found even before Abrikosov, just the other way around. It is known that the large-U limit of the Hubbard model at half-filling gives a antiferromagnetic Heisenberg model. In the perturbation theory one finds the exact representation of Spins in terms of fermionic ladder operators given above.

2.1.1 Hilbert space and un-physical states

The mapping 19 is exact on the operator level, meaning, that the eigenspaces of S_i^2 and S_i^α remain unchanged.

The local Hilbert space, however, gets enlarged. We drop the site-indices from now on. The spin-1/2 Hilbert space is of course a two-dimensional Hilbert space

$$\mathcal{H}_{S=\frac{1}{2}} = \text{span}\left(|\uparrow\rangle, |\downarrow\rangle\right) \tag{20}$$

The two-fermion Hilbert space on the other hand is four-dimenstional

$$\mathcal{H}_{\text{fermions}} = \text{span}\left(|00\rangle, |10\rangle, |01\rangle, |11\rangle\right). \tag{21}$$

The idea of 19 is to identify $\mathcal{H}_{S=\frac{1}{2}}$ with the one-particle sector of $\mathcal{H}_{\text{fermions}}$:

$$\left(n_{i\uparrow} + n_{i\downarrow}\right)|\psi\rangle \stackrel{!}{=} |\psi\rangle \;\; \forall i. \tag{22}$$

This leaves us with two un-physical states: $|00\rangle$ and $|11\rangle$.

All spin-operators S^α act trivially on these, which is why everything goes through smoothly on the operator level. Therefore, as long as the Hamiltonian only consists of spin operators, the un-physical states do not influence any results (apart from normalization, maybe).

This can not be garuanteed anymore when treating the system perturbatively using for example a mean-field theory. For this reason we introduce the constraint

$$\langle\psi| n_{i\downarrow} + n_{i\uparrow} |\psi\rangle = 1 \; \forall i. \tag{23}$$

which is a constraint on the state $|\psi\rangle$. Acknowledge, that this constraint is weaker then $n_i = 1 \; \forall i$, for example

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right) \quad \Rightarrow \quad \langle\psi| n_\uparrow + n_\downarrow |\psi\rangle = 1 \quad \text{but} \quad \left(n_{i\uparrow} + n_{i\downarrow}\right)|\psi\rangle = \sqrt{2}\,|11\rangle \neq |\psi\rangle. \tag{24}$$

Obviously all normalized states that fulfill 22 also comply with 23.

2.1.2 global SU(2) symmetrie

Many important models, e.g. exchange models, have spin rotation symmetrie. The symmetrie group of spin-rotations is SU(2). Rewriting 19 in Matrix notation (for S=1/2, without position indices):

$$\vec{S} = \frac{1}{2}\begin{pmatrix} c^\dagger_\uparrow & c^\dagger_\downarrow \end{pmatrix} \vec{\sigma} \begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix} \tag{25}$$

the spin rotation can be represented by $\vec{\sigma} \mapsto V^{-1}\vec{\sigma}V$, with V an element of SU(2). The task at hand is to determine, how the spin rotation acts on the ladder operators. One may reinterpret

$$\vec{S} \to \frac{1}{2}\begin{pmatrix} c_\uparrow^\dagger & c_\downarrow^\dagger \end{pmatrix} V^{-1}\vec{\sigma}V \begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix} \tag{26}$$

as

$$\begin{pmatrix} c_\uparrow^\dagger & c_\downarrow^\dagger \end{pmatrix} \to \begin{pmatrix} c_\uparrow^\dagger & c_\downarrow^\dagger \end{pmatrix} V^{-1} \quad \begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix} \to V \begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix} \quad \vec{\sigma} \to \vec{\sigma}. \tag{27}$$

$c_\downarrow$, $c_\uparrow$ transform as a SU(2) doublet under left multiplication by an SU(2) matrix. To construct the conjugate doublet, which transforms in the same manner, consider

$$\begin{pmatrix} c_\uparrow^\dagger \\ c_\downarrow^\dagger \end{pmatrix} \to \left[\begin{pmatrix} c_\uparrow^\dagger & c_\downarrow^\dagger \end{pmatrix} V^{-1}\right]^T = \left(V^{-1}\right)^T \begin{pmatrix} c_\uparrow^\dagger \\ c_\uparrow^\dagger \end{pmatrix} = \left(V^{-1}\right)^T \sigma_3\sigma_3 \begin{pmatrix} c_\uparrow^\dagger \\ c_\uparrow^\dagger \end{pmatrix} \tag{28}$$

To recover V^{-1}, multiply both sides by σ_3 from the left and use $V^{-1} = V^H = \sigma_3 V^T \sigma_3$ (σ_3 represents the complex conjugation in SU(2) better formulation?).

$$\Rightarrow \sigma_3 \begin{pmatrix} c_\uparrow^\dagger \\ c_\downarrow^\dagger \end{pmatrix} \to V\sigma_3 \begin{pmatrix} c_\uparrow^\dagger \\ c_\downarrow^\dagger \end{pmatrix} \tag{29}$$

.
Hence, the two conjugate doublets are

$$\begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} c_\uparrow^\dagger \\ -c_\downarrow^\dagger \end{pmatrix} \tag{30}$$

.
It is convention two write these two doublets in a matrix:

$$\psi = \begin{pmatrix} c_\uparrow & c_\downarrow \\ c_\downarrow^\dagger & -c_\uparrow^\dagger \end{pmatrix}, \tag{31}$$

the spin rotation is then $\psi \mapsto \psi V$. Hence, the combination $\psi_i\psi_j^\dagger$ is manifestly invariant. This reveals two rotationally invariant operators χ_{ij} and η_{ij} :

$$\psi\psi^\dagger = \begin{pmatrix} c_{i\uparrow}^\dagger c_{j\uparrow}^\dagger + c_{i\downarrow}c_{j\downarrow}^\dagger & c_{i\uparrow}c_{j\downarrow} - c_{i\downarrow}c_{j\uparrow} \\ c_{i\downarrow}^\dagger c_{j\uparrow}^\dagger - c_{i\uparrow}^\dagger c_{j\downarrow}^\dagger & c_{i\downarrow}^\dagger c_{j\downarrow} + c_{i\uparrow}^\dagger c_{j\uparrow} \end{pmatrix} \equiv \begin{pmatrix} -\chi_{ij}^\dagger & -\eta_{ij}^\dagger \\ -\eta_{ij} & \chi_{ij} \end{pmatrix} \tag{32}$$

The spint operators can be expressed in manifestly rotationally-invariant form:

$$S_i^\alpha = \frac{1}{2}\mathrm{Tr}\left[\psi_i^\dagger\psi_i\left(\sigma\right)^T\right] \tag{33}$$

2.1.3 local SU(2) symmetrie

Consider

$$S_+\left|\downarrow\right\rangle = \left|\uparrow\right\rangle, \tag{34}$$

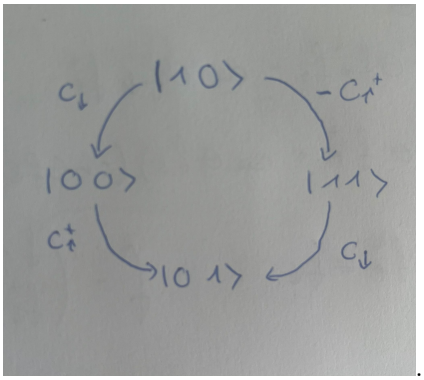
which is an operation on $\mathcal{H}_{S=\frac{1}{2}}$. When translating this into $\mathcal{H}_{\text{fermions}}$:

$$S_+ = \frac{1}{2}(S_x + iS_y) \to c_\uparrow^\dagger c_\downarrow, \tag{35}$$

one uncovers a redundancy in the description:

$$c_\uparrow^\dagger c_\downarrow = \sin^2(\theta)c_\uparrow^\dagger c_\downarrow + \cos^2(\theta)c_\uparrow^\dagger c_\downarrow = \sin^2(\theta)c_\uparrow^\dagger c_\downarrow - \cos^2(\theta)c_\downarrow c_\uparrow^\dagger. \tag{36}$$

On the level of local hilbert spaces, this can be understood as different walks connecting the same initial and final states (hence, an artifact of the enlargement).



The physical interpretation is that $c_\downarrow$ and $-c_\uparrow^\dagger$ hace the same action: they decrease S_z by unit. Suddenly, we have a gauge-structure at our hands. The next questions come naturally: what are general gauge transformations and what are gauge invariants? We conbine the four operators into two doublets:

$$\begin{pmatrix} c_\downarrow \\ -c_\uparrow^\dagger \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} c_\uparrow \\ c_\downarrow^\dagger \end{pmatrix}, \tag{37}$$

which transform by left multiplication by a matrix (the gauge transformation). These doublets again form 36, From the representation of the spin-operators it is obvious that the gauge transformatoins have to be from $SU(2)$.

3. Mechanics of Continua

3.1 Kinematics

3.1.1 Notation and Coordinates

I will skip over the very technical mathmatical description of the basic objects (like bodies and points) and skip ahead to the interesting stuff.

It is convention to use large latin letters for quantities in the reference configuration and small latin letters for quantities of a particular configuration. One might also refer to the reference configuratoin as the initial configuration.

For example: the position of a particle in the reference configuration is called X, while it is called x in a particular configuration.

This notation gets inherited by all objects in the theory through the variabeles they depend on.

3.1.2 Eulerian and Lagrangian description

When describing how the configuration of a body changes over time it can be advantageous to describe different processes from the points of views of different observers. It comes as no surprise that the choice of description does not change the physics but is merely a matter of convencience. There are two canonical descriptions to choose from;

In the **Eulerian description** (or spatial description) $X(x,t)$ is a function of t and x.

The intutive picture is that of an observer that observes a certain point x in space. Like a hiker resting next to stream, observing it as it flows by. If they encounter particle P at time t (at their point of observation x), they can infer the point $X_P(x,t)$, where P would have been in the reference configuration.

In this sense x is the only free variable besides t. Therefore, quantities in the Eulerian picture are always denoted by small latin letters.

In the **Lagrangian description** (or material description) $x(X,t)$ is a function of t and X.

In this case the observer follows a particle (or a collection of particles, called a region) as it evolves over time. Imagine the hiker built a raft and rides it down the afformentioned stream (assuming the rafts velocity matches that of the stream). Knowing the point X where they started rafting they can reason where they will end up at time t; the point being $x(X,t)$.

Hence, here X and t are the free coordinates. All material quantities are denoted by capital greek letters.

3.1.3 Trajectories and Displacment Field

The object conntecting the afformentioned descriptions is the **trajectorie** $\chi(X,t)$:

$$x = \chi(X,t) \quad \Leftrightarrow \quad X = \chi^{-1}(x,t). \tag{38}$$

It maps out where a observed particle that started at X will be at time t. Obviously, χ itself is part of the Lagrangian description (χ^{-1} belongs to the Eulerian description). A function $f(x,t)$ in Eulerian description can be transformed according to:

$$f(x,t) \mapsto F(X,t) = f(\chi(X,t),t), \tag{39}$$

where F is now in the Lagrangian description. Though f and F are conceptually equivalent they are, mathematically speaking, two different objects.

An example for such a fuunction is the **displacement field**:

$$\begin{aligned} U(X,t) &= X - x(X,t) \\ u(x,t) &= X(x,t) - x \end{aligned} \tag{40}$$

$$U(X,t) \mapsto u(x,t) = U(\chi^{-1}(x,t),t) \tag{41}$$

The displacement field describes how much a particle is displaced compared to the reference configuration.

3.1.4 local changes and Deformation Gradient

Part II
main part

4. Triangular Lattice

4.1 Construction of Hamiltonian

rename this section to something like model?

4.1.1 Construction of the Hamiltonian

The Hamiltonian is

$$\mathcal{H} = \sum_{\langle ij \rangle} J_{ij} \left(\vec{S}_i \cdot \vec{S}_j + \kappa \left(\vec{S}_i \cdot \vec{S}_j \right)^2 \right) \tag{42}$$

Using the Abrikosov representation of the spin in terms of fermions:

$$\vec{S}_i = \sum_{\alpha\beta} f^\dagger_{i\alpha} \frac{\vec{\sigma}}{2} f_{i\beta} \tag{43}$$

$$\begin{aligned} \Rightarrow \vec{S}_i \cdot \vec{S}_j &= \frac{1}{4} \sum_{\alpha\beta\gamma\delta} f^\dagger_{i\alpha} f_{i\beta} f^\dagger_{j\gamma} f_{j\delta} \underbrace{\vec{\sigma}_{\alpha\beta} \vec{\sigma}_{\gamma\delta}}_{=\delta_{\alpha\delta}\delta_{\beta\gamma} - \delta_{\alpha\beta}\delta_{\gamma\delta}} \\ &= \frac{1}{4} \sum_{\alpha\beta} 2 f^\dagger_{i\alpha} f_{i\beta} f^\dagger_{j\beta} f_{j\alpha} - f^\dagger_{i\alpha} f_{i\alpha} f^\dagger_{j\beta} f_{j\beta} \end{aligned}$$

Move the last annihilation operator in the first term to the second postion: use $f^\dagger_{j\beta} f_{j\alpha} = \delta_{\alpha\beta} - f_{j\alpha} f^\dagger_{j\beta}$ and acquire 1 minus signs. After that bring the other annihilation operator to the last position which gives another minus.

$$\vec{S}_i \cdot \vec{S}_j = \frac{1}{4} \sum_{\alpha\beta} -2 f^\dagger_{i\alpha} f_{j\alpha} f^\dagger_{j\beta} f_{j\beta} + 2 f_{i\alpha} f_{i\beta} \delta_{\alpha\beta} - f^\dagger_{i\alpha} f_{i\alpha} f^\dagger_{j\beta} f_{j\beta} \tag{44}$$

The second term is n_i and the third is $n_i n_j$. For the first term I introduce the shorthand

$$\chi_{ij} = \frac{1}{2} \sum_{\alpha} f^\dagger_{i\alpha} f_{j\alpha} \tag{45}$$

$$\vec{S}_i \cdot \vec{S}_j = -2 \chi_{ij} \chi_{ji} + \frac{1}{2} n_i - \frac{1}{4} n_i n_j \tag{46}$$

Using the constaint $n_i = n_j = 1$, the final expression is

$$\vec{S}_i \cdot \vec{S}_j = -2 \chi_{ij} \chi_{ji} + \frac{1}{4} \tag{47}$$

Relevant for the Hamiltonian is also the term:

$$\Rightarrow \left(\vec{S}_i \cdot \vec{S}_j \right)^2 = \frac{1}{16} - \chi_{ij} \chi_{ji} + 4 \left(\chi_{ij} \chi_{ji} \right)^2 \tag{48}$$

The mean field decoupling works like

$$\chi_{ij}\chi_{ji} \approx \langle\chi_{ij}\rangle\chi_{ji} + \langle\chi_{ji}\rangle\chi_{ij} - |\langle\chi_{ij}\rangle|^2 \tag{49}$$

$$(\chi_{ij}\chi_{ji})^2 \approx 2|\langle\chi_{ij}\rangle|^2\langle\chi_{ij}\rangle\chi_{ji} + 2|\langle\chi_{ij}\rangle|^2\langle\chi_{ji}\rangle\chi_{ij} - 3|\langle\chi_{ij}\rangle|^4 \tag{50}$$

$$\Rightarrow \vec{S}_i \cdot \vec{S}_j \approx -2\langle\chi_{ij}\rangle\chi_{ji} + \text{h.c.} + 2|\langle\chi_{ij}\rangle|^2 + \frac{1}{4} \tag{51}$$

$$\Rightarrow \left(\vec{S}_i \cdot \vec{S}_j\right)^2 \approx -\langle\chi_{ij}\rangle\chi_{ji} + 8|\langle\chi_{ij}\rangle|^2\langle\chi_{ij}\rangle\chi_{ji} + \text{h.c.} + |\langle\chi_{ij}\rangle|^2 - 12|\langle\chi_{ij}\rangle|^4 + \frac{1}{16} \tag{52}$$

Maybe comment of the fact that the mean-field parameters know about the fermi sea and thereby about many states. In this sense the states do speak with each other. Bringing everything together:

$$\begin{aligned}\mathcal{H} \approx \sum_{\langle ij\rangle} J_{ij}\Big[&\left(-1-\kappa/2+4\kappa|\langle\chi_{ij}\rangle|^2\right)\langle\chi_{ij}\rangle\left(f^\dagger_{j\uparrow}f_{i\uparrow}+f^\dagger_{j\downarrow}f_{i\downarrow}\right)+\text{h.c}\\&+\left(2+\kappa-12\kappa|\langle\chi_{ij}\rangle|^2\right)|\langle\chi_{ij}\rangle|^2+\frac{1}{4}+\frac{\kappa}{16}\Big]\end{aligned} \tag{53}$$

It is sometimes handy to define

$$\tilde{J}_{ij} = J_{ij}\left(1+\kappa/2\right), \quad \tilde{\kappa} = \kappa/(1+\kappa/2) \tag{54}$$

I drop the constant:

$$\begin{aligned}\mathcal{H} \approx \sum_{\langle ij\rangle} \tilde{J}_{ij}\Big[&-\left(1-4\tilde{\kappa}|\langle\chi_{ij}\rangle|^2\right)\langle\chi_{ij}\rangle\left(f^\dagger_{j\uparrow}f_{i\uparrow}+f^\dagger_{j\downarrow}f_{i\downarrow}\right)+\text{h.c}\\&+2\left(1-6\tilde{\kappa}|\langle\chi_{ij}\rangle|^2\right)|\langle\chi_{ij}\rangle|^2\Big]\end{aligned} \tag{55}$$

This complies with the result obtained here [**?**]. Hence, the mean field produces the effective hopping amplitude t^{MF}:

$$t^{\text{MF}}_{ji} = -\tilde{J}_{ij}\langle\chi_{ij}\rangle\left(1-4\tilde{\kappa}|\langle\chi_{ij}\rangle|^2\right) = -J_{ij}\langle\chi_{ij}\rangle\left(1+\kappa/2-4\kappa|\langle\chi_{ij}\rangle|^2\right). \tag{56}$$

$$\Rightarrow \mathcal{H} \approx \sum_{\langle ij\rangle} t^{\text{MF}}_{ji}\left(f^\dagger_{j\uparrow}f_{i\uparrow}+f^\dagger_{j\downarrow}f_{i\downarrow}\right)+\text{const.} \tag{57}$$

This is pretty interesting: the hopping-amplitude $i \to j$ is directly proportional to the hopping-amplitude $j \to i$. Here are a few ideas how to make use/discuss this insight:

- Maybe one can also make the argument that we start with a "symmetric" model, i.e. a model wohtout internal sense of direction, a feature that is now preserved in the Mean field Hamiltonian.
- Or maybe, even more suffisticated, one could make argument using strauctures of the spin-singlet states.
- Maybe one can use this fact to siplify the Mean-field equation?

4.1.2 Continuity equation and probability current

The probability (/particle) density on site k is

$$n_k = \sum_\sigma f^\dagger_{k\sigma} f_{k\sigma}. \tag{58}$$

Now I derive the corresponding probability current density:

$$\frac{\partial n_k}{\partial t} = -i \sum_\sigma \left([f^\dagger_{k\sigma}, H] f_{k\sigma} + f^\dagger_{k\sigma} [f_{k\sigma}, H] \right) \tag{59}$$

$$\begin{aligned} [f^\dagger_{k\sigma}, H] f_{k\sigma} &= -\sum_\delta t_{k+\delta;k} f^\dagger_{k+\delta;\sigma} f_{k\sigma} - \mu_k f^\dagger_{k\sigma} f_{k\sigma} \\ f^\dagger_{k\sigma} [f_{k\sigma}, H] &= \sum_\delta t_{k;k+\delta} f^\dagger_{k\sigma} f_{k+\delta;\sigma} + \mu f^\dagger_{k\sigma} f_{k\sigma} \end{aligned} \tag{60}$$

$$\Rightarrow \frac{\partial n_k}{\partial t} = -\sum_\delta i \left(\sum_\sigma t_{k,k+\delta} f^\dagger_{k\sigma} f_{k+\delta;\sigma} - t_{k+\delta,k} f^\dagger_{k+\delta;\sigma} f_{k\sigma} \right) \tag{61}$$

Define

$$J_{k\to k+\delta} \equiv i \left(\sum_\sigma t_{k,k+\delta} f^\dagger_{k\sigma} f_{k+\delta;\sigma} - t_{k+\delta,k} f^\dagger_{k+\delta;\sigma} f_{k\sigma} \right) \tag{62}$$

$J_{k\to k+\delta}$ is the (directed!) current density along the bond $\vec{R}_k \to \vec{R}_{k+\delta}$. The discrete (/lattice) continuity equation reads

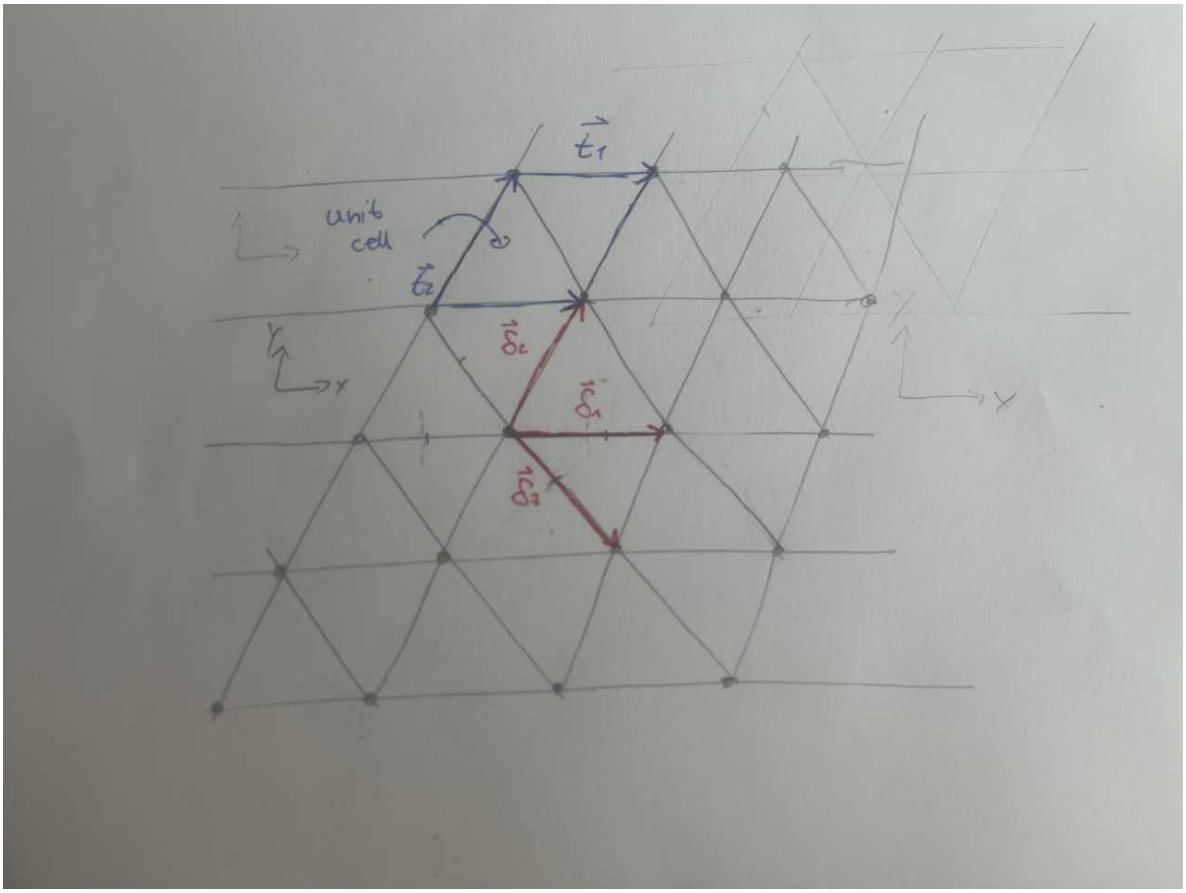
$$\frac{\partial n_k}{\partial t} = -\sum_\delta J_{k\to k+\delta}. \tag{63}$$

Something interesting happens when calculating $\langle J_{k\to k+\delta} \rangle$:

$$\begin{aligned} \langle J_{k\to k+\delta} \rangle &= i \left(t_{k,k+\delta} \left\langle \sum_\sigma f^\dagger_{k\sigma} f_{k+\delta;\sigma} \right\rangle - t_{k+\delta,k} \left\langle \sum_\sigma f^\dagger_{k+\delta;\sigma} f_{k\sigma} \right\rangle \right) \\ &= 2i \left(t_{k,k+\delta} \langle \chi_{k,k+\delta} \rangle - t_{k+\delta,k} \langle \chi_{k+\delta,k} \rangle \right) \\ &= -2i \tilde{J}_{k;k+\delta} \left(1 - 4\tilde{\kappa} |\langle \chi_{k,k+\delta} \rangle|^2 \right) \left(\langle \chi_{k+\delta;k} \rangle \langle \chi_{k;k+\delta} \rangle - \langle \chi_{k;k+\delta} \rangle \langle \chi_{k+\delta;k} \rangle \right) \\ &= 0. \end{aligned} \tag{64}$$

Apparently, the current on each bond vanishes in the ground state by construction. This still leaves the possibilitie that single states have non-zero probability currents. However, it remains to investigate how this is possible for e.g. states with a singular (topologically protected) edge mode. One would guess that this feature would lead to a non-zero current in the ground state.

4.2 0-0 and π-π States



This system is a simple triangular Bravais lattice with primitve translations

$$\vec{a}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{a}_2 = \begin{pmatrix} 1/2 \\ \sqrt{3}/2 \end{pmatrix}. \tag{65}$$

4.2.1 Pedestrian method

The tight-binding Hamiltonian is:

$$\begin{aligned} H &= \sum_{i,\delta,\sigma} \frac{t}{2} \left(f^\dagger_{i\sigma} f_{i+\delta\,\sigma} + \text{h.c.} \right) + \mu \sum_i \left(f^\dagger_{i\uparrow} f_{i\uparrow} + f^\dagger_{i\downarrow} f_{i\downarrow} - 1 \right) + \text{const.} \\ &= \sum_{i,\delta,\sigma} \frac{t}{2} \left(f^\dagger_{i\sigma} f_{i+\delta\,\sigma} + \text{h.c.} \right) + \mu \sum_{i,\sigma} f^\dagger_{i\sigma} f_{i\sigma} - \mu N + \text{const.}. \end{aligned} \tag{66}$$

i runs over sites sites and δ over nearest neighbours of a generic site. I include the absolute shift for the sake of consistency. The factor $1/2$ is also included for consistency. The constant is a product of mean field theory and reads

$$\frac{1}{2} \sum_{i,\delta} 2\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta} |^2 \rangle \right) |\langle \chi_{i,i+\delta} \rangle|^2 = 6N \left(\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta} \rangle|^2 \rangle \right) |\langle \chi_{i,i+\delta} \rangle|^2 \right) \tag{67}$$

The second step can be understood in at least two ways; a) each site has 6 neighbours, or b) there are bond per primitive unit cell 1 of which is completely inside the unit cell and contributes a 2 and the

other 4 shared between 2 unit cells thereby contributing a 1. All bonds are equivalent of course. The afformentioned six nearest neighbours lie at the (relative) positions

$$\vec{\delta_1} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \; \vec{\delta_2} = \begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix} \; \vec{\delta_3} = \begin{pmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{pmatrix}$$

and $-\vec{\delta_1}$, $-\vec{\delta_2}$, $-\vec{\delta_3}$. I will start with the $0-0$ state. To advance, do a Fourier transform:

$$f_{i\sigma} = \frac{1}{\sqrt{N}} \sum_k e^{i\vec{k}\vec{R}_i} c_{k\sigma}. \tag{68}$$

$$\begin{aligned} \Rightarrow H &= \sum_{q,\sigma} \frac{t}{2} \left(c^\dagger_{q\sigma} c_{q\sigma} \left(\sum_\delta e^{i\vec{q}\vec{\delta}} + \mu \right) + \text{h.c.} \right) - \mu N \\ &= \sum_{q,\sigma} \left(t \sum_\delta \cos\left(\vec{q}\vec{\delta}\right) + \mu \right) c^\dagger_{q\sigma} c_{q\sigma} - \mu N \end{aligned} \tag{69}$$

Using first the symmetrie of the cosine and then the sum identity for the cosine the dispersion reads

$$\begin{aligned} \varepsilon(\vec{q}) &= 2t \left[\cos(q_x) + \cos\left(\frac{1}{2}(q_x + \sqrt{3} q_y) \right) + \cos\left(\frac{1}{2}(q_x - \sqrt{3} q_y) \right) \right] + \mu \\ &= 2t \left[\cos(q_x) + 2\cos\left(\frac{q_x}{2} \right) \cos\left(\frac{\sqrt{3} q_y}{2} \right) \right] + \mu \end{aligned} \tag{70}$$

features:

- $\varepsilon(\vec{q}) \in [-3t, 6t] + \mu \Rightarrow \mu \in [-6t, 3t]$
- symmetries $q_x \to -q_x$ and $q_y \to -q_y$ (+ should have some rotational symmetrie and (thereby associated) inflections along other axes) could check this
- periodicity: 4π in q_x, $(4\pi)/\sqrt{3}$ in q_y

The mean field parameters are

$$\langle \chi_{ij} \rangle = \frac{1}{2} \langle f^\dagger_{i\uparrow} f_{j\uparrow} + f^\dagger_{i\downarrow} f_{j\downarrow} \rangle = \langle f^\dagger_{i\uparrow} f_{j\uparrow} \rangle \tag{71}$$

Simplify the expression by Fourier transform:

$$\begin{aligned} \langle \chi_{ij} \rangle &= \frac{1}{N} \sum_{q,p} e^{i(\vec{q}\vec{R}_j - \vec{k}\vec{R}_i)} \langle c^\dagger_p c_q \rangle \\ &= \frac{1}{N} \sum e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} f(\varepsilon(\vec{q})) \end{aligned} \tag{72}$$

Take the thermodynamic limit $\sum_k \to (N\Omega)/(2\pi)^2 \int dk$ (Ω is the unit cell volume).

$$\Rightarrow \langle \chi_{ij} \rangle = \Omega \int_{1.\text{BZ}} \frac{d^2 q}{(2\pi)^2} e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} f(\varepsilon(\vec{q})) \tag{73}$$

It looks like the model allows for different χ_{ij} in each direction $\vec{R}_j - \vec{R}_i$ (there are 3 of those)?

From this expression, recover the on-site particle number by setting $i = j$ and accounting for the factor $1/2$ in the definition of χ_{ij}:

$$\langle N_i \rangle = 2\Omega \int \frac{d^2q}{(2\pi)^2} f(\varepsilon(\vec{q})). \tag{74}$$

By construction the particle number is uniform.
Some elaborations on the integral domain follow. The primite translations of the reciprocal lattice are

$$\vec{b}_1 = 2\pi \begin{pmatrix} 0 \\ 2/\sqrt{3} \end{pmatrix}, \quad \vec{b}_2 = 2\pi \begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix} \tag{75}$$

I do a basis transformation onto thos axes defined by

$$\begin{pmatrix} k_x \\ k_y \end{pmatrix} = \begin{pmatrix} 1/2 & \sqrt{3}/2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} q_x \\ q_y \end{pmatrix} \tag{76}$$

$$\Leftrightarrow \begin{pmatrix} q_x \\ q_y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 2/\sqrt{3} & -1/\sqrt{3} \end{pmatrix} \begin{pmatrix} k_x \\ k_y \end{pmatrix} \tag{77}$$

$$\Rightarrow \det J = -2/\sqrt{3} \tag{78}$$

$$\Rightarrow \varepsilon(\vec{k}) = 2t\left(\cos(k_x) + \cos(k_y) + \cos(k_x - k_y)\right) + \mu \tag{79}$$

The first Brillouin zone is $\{k_x, k_y\} \in [-\pi, \pi]^2$. The unit cell volume is $\Omega = \sqrt{3}/2$. Consistency check: $|\det(\mathrm{J})| = \Omega^{-1}$, since the substitution simply transforms the unti cell to a square of unit volume. So the integrals that have to be evaluated are of the form

$$\langle N_i \rangle = 2 \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} f(\varepsilon(\vec{k}), \mu), \tag{80}$$

$$\langle \chi_{ij} \rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\Big[i\left(k_y, (1/\sqrt{3})(2k_x - k_y)\right) \cdot (\vec{R}_j - \vec{R}_i)\Big] f(\varepsilon(\vec{k}), \mu) \tag{81}$$

Check that under a particle hole transformation chi changes sign, sign of t is doing just that and this brings the *pi*-flux phase.
The condition is $\langle N_i \rangle \stackrel{!}{=} 1 \; \forall i$. By setting setting $t = -1$, the chemical potential is numerically determined in units of t using mathematica. I choose $t = -1$ because I anticipate t to be negative for a positive χ. (This may not be true depending on κ and χ, but as long as it is self-consistent evrything is fine.) For the numerical integration I used a Fermi-function with a small temperature ($\beta = 200t$) instead of a step-function, since the ladder is not analytical and causes problems for the numerical algorithm. The result is $\mu = -0.834737\ t$. The bond-parameters are calulcated the same way, the result is $\langle \chi_{ij} \rangle = 0.164713$.

Now, it is possible to express μ only in units of J. Use the formula

$$t_{ij} = -J\langle \chi_{ij} \rangle^* (1 + \kappa/2 - 4\kappa\, |\langle \chi_{ij} \rangle|^2). \tag{82}$$

Set $\kappa = 10$:

$$t_{ij} = -0.8095\ J \tag{83}$$

$$\mu = -0.67574\ J \tag{84}$$

$$\langle \chi_{ij} \rangle = 0.164713 \tag{85}$$

The energy density (that is also the free energy desnity at $T = 0$) is

$$\frac{\langle H\rangle}{N} = 2 \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}\varepsilon(\vec{k})f(\varepsilon(\vec{k})) - \mu + 6\tilde{J}\left(1 - 6\tilde{\kappa}|\langle\chi_{i,i+\delta}\rangle|^2\right)|\langle\chi_{i,i+\delta}\rangle|^2 \; = -0.888304\,J \tag{86}$$

Observtaion; the chemical potential drops out of the Free energy.

One can also directly calculate results for the state with all fluxes π by performing a particle-hole-transformation:

$$t \to -t$$
$$\mu \to -\mu \tag{87}$$
$$\Rightarrow \varepsilon(\vec{k}) \to -\varepsilon(\vec{k}) \tag{88}$$
$$\Rightarrow f(\varepsilon(\vec{k})) \to 1 - f(\varepsilon(\vec{k}) \tag{89}$$

It can be shown that

$$\chi \to -\chi. \tag{90}$$

The result is:

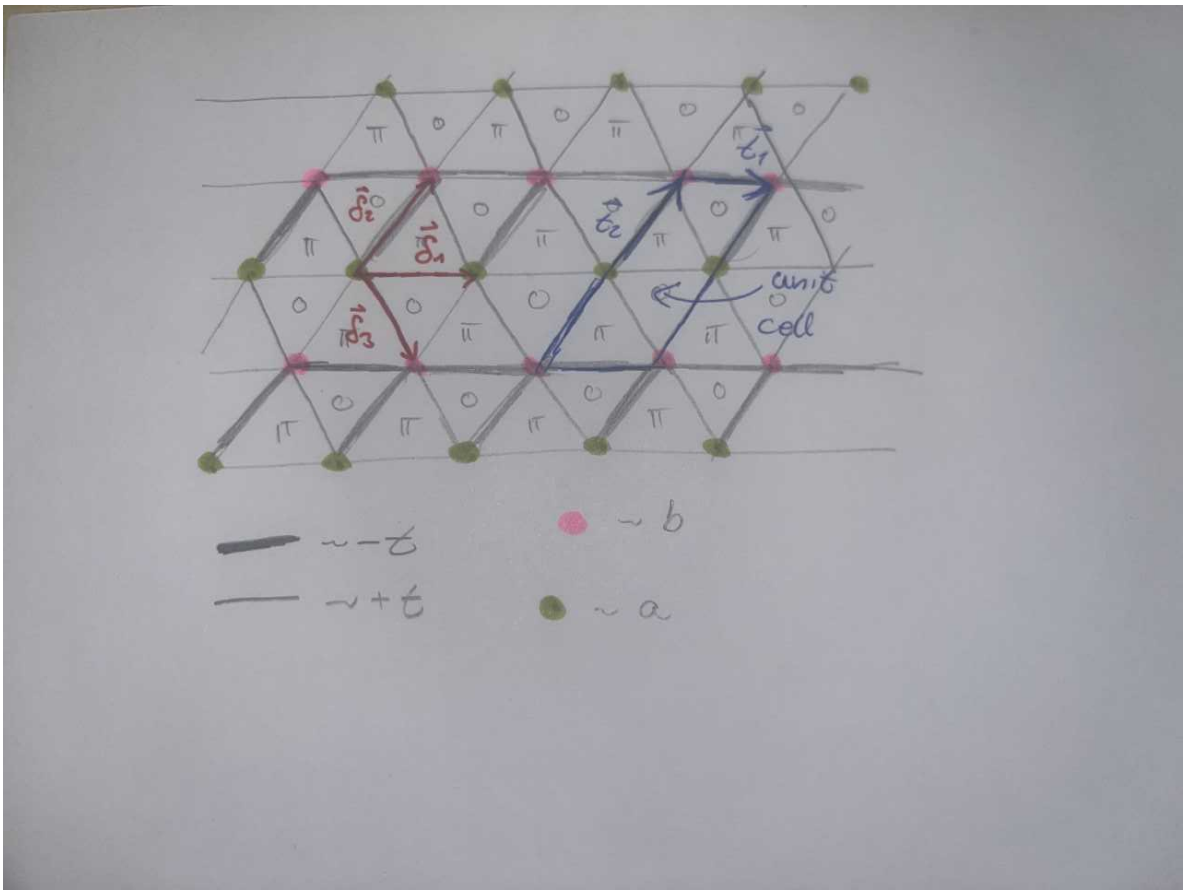
$$\frac{\langle H\rangle_\pi}{N} = -0.888302\,J \tag{91}$$

4.2.2 Pedro's method

The longer explenation for this method is in the next section. Here I will only say that the free energy is the same except for γ, which gets replaced by a factor ϕ (as mentioned above the chemical potentioal does not contribute to the free energy):

4.3 0-π and π-0 States

4.3.1 pedestrian method



The system is a triangular bravais lattice with a two-atomic basis. The primitve translations are

$$\vec{a}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{a}_2 = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \tag{92}$$

Hence, two types of lattice sites exist, a and b (see sketch). Also χ_{ij}, and thereby t_{ij} can change sign (see sketch). The Hamiltonian is

$$H = \frac{1}{2} \sum_{\langle i,j \rangle, \sigma} \left(t a^\dagger_{i\sigma} a_{j\sigma} - t b^\dagger_{i\sigma} b_{j\sigma} + t^{(ab)}_{ij} a^\dagger_{i\sigma} b_{j\sigma} + \text{h.c.} \right) + \mu_a \sum_{i,\sigma} \left(a^\dagger_{i\sigma} a_{i\sigma} - 1 \right) + \mu_b \sum_{i,\sigma} \left(b^\dagger_{i\sigma} b_{i\sigma} - 1 \right) + \text{const.} \tag{93}$$

$$\begin{aligned} H = \frac{t}{2} \sum_{i\sigma} \Big(& a^\dagger_{i\sigma} a_{i+\delta_1;\sigma} + a^\dagger_{i\sigma} a_{i-\delta_1;\sigma} - a^\dagger_{i\sigma} b_{i+\delta_2;\sigma} + a^\dagger_{i\sigma} b_{i-\delta_2;\sigma} + a^\dagger_{i\sigma} b_{i+\delta_3;\sigma} + a^\dagger_{i\sigma} b_{i-\delta_3;\sigma} \\ & - b^\dagger_{i\sigma} b_{i+\delta_1;\sigma} - b^\dagger_{i\sigma} b_{i-\delta_1;\sigma} + b^\dagger_{i\sigma} a_{i+\delta_2;\sigma} - b^\dagger_{i\sigma} a_{i-\delta_2;\sigma} + b^\dagger_{i\sigma} a_{i+\delta_3;\sigma} + b^\dagger_{i\sigma} a^\dagger_{i-\delta_3;\sigma} \\ & + \mu_a a^\dagger_{i\sigma} a_{i\sigma} + \mu_b b^\dagger_{i\sigma} b_{i\sigma} \Big) - \frac{N(\mu_a + \mu_b)}{2} \end{aligned} \tag{94}$$

The site index notation is not ideal, as a and b sites by definition cant have the same site indices. The constant is the same as for the 0-flux case, provided that the absolute value of $\chi_{i,i+\delta}$ is uniform.

$$\frac{1}{2}\sum_{i,\delta} 2\tilde{J}\left(1-6|\langle\chi_{i,i+\delta}|^2\rangle\right)|\langle\chi_{i,i+\delta}\rangle|^2 = 6N\tilde{J}\left(1-6\tilde{\kappa}|\langle\chi_{i,i+\delta}\rangle|^2\right)|\langle\chi_{i,i+\delta}\rangle|^2 \tag{95}$$

This is not quite wright yet. The Fourier transformation has to be modified bc there are only N/2 of a and b-sites respectively

$$a_{i\sigma} = \sqrt{\frac{2}{N}}\sum_k e^{i\vec{k}\vec{R}_i} a_{k\sigma}, \tag{96}$$

and the same for $b_{i\sigma}$. The Hamiltonian (without the constants) becomes

$$H = \sum_{q,\sigma}\begin{pmatrix} a^\dagger_{q\sigma} & b^\dagger_{q\sigma}\end{pmatrix}\begin{pmatrix} 2t\cos\left(\vec{q}\vec{\delta}_1\right)+\mu_a & 2t\cos\left(\vec{q}\vec{\delta}_3\right)-2it\sin\left(\vec{q}\vec{\delta}_2\right) \\ 2t\cos\left(\vec{q}\vec{\delta}_3\right)+2it\sin\left(\vec{q}\vec{\delta}_2\right) & -2t\cos\left(\vec{q}\vec{\delta}_1\right)+\mu_b\end{pmatrix}\begin{pmatrix} a_{q\sigma} \\ b_{q\sigma}\end{pmatrix}. \tag{97}$$

The primitive translations of the reciprocal lattice are

$$\vec{b}_1 = 2\pi\begin{pmatrix}0\\1/\sqrt{3}\end{pmatrix}, \quad \vec{b}_2 = 2\pi\begin{pmatrix}1\\-1/\sqrt{3}\end{pmatrix} \tag{98}$$

.
The suitable transformation is

$$\begin{pmatrix}k_x\\k_y\end{pmatrix} = \begin{pmatrix}1 & \sqrt{3}\\ m1 & 0\end{pmatrix}\begin{pmatrix}q_x\\q_y\end{pmatrix} \tag{99}$$

$$\Leftrightarrow \begin{pmatrix}q_x\\q_y\end{pmatrix} = \begin{pmatrix}0 & 1\\ 1/\sqrt{3} & -1/\sqrt{3}\end{pmatrix}\begin{pmatrix}k_x\\k_y\end{pmatrix} \tag{100}$$

$$\Rightarrow \det(J) = 1/\sqrt{3} \tag{101}$$

This yields

$$\cos\left(\vec{q}\vec{\delta}_1\right) \to \cos(k_y)$$

$$\sin\left(\vec{q}\vec{\delta}_2\right) \to \sin(k_x/2) \tag{102}$$

$$\cos\left(\vec{q}\vec{\delta}_3\right) \to \cos(k_y - k_x/2) \tag{103}$$

$$\varepsilon(\vec{k})_\pm = \mu_a+\mu_b\pm\sqrt{(\mu_a-\mu_b)^2+2t\cos(k_y)(\mu_a-\mu_b)+4t^2\cos^2(k_y)+4t^2\cos^2(k_y-k_x/2)+4t^2\sin^2(k_x/2)} \tag{104}$$

Apparently, half filling is achieved globally, iff $\mu_a = -\mu_b \equiv \mu$. (The spectrum is symmetric around $\varepsilon = 0$ in this case.) In order to fulfill the local constraint one still has to calculate the local expectation values. It would be weird however, if the chemical potential differed between a and b sites (-¿ charge density waves?).

$$\Rightarrow \varepsilon(\vec{k})_\pm = \pm 2|t|\sqrt{(\mu/(2t)+\cos(k_y))^2+\cos^2(k_y-k_x/2)+\sin^2(k_x/2)} \tag{105}$$

It is correct that ε only depends on the absolute value of t, it looks like this would completely neutrilise the particle-hole transformation.The periodicity is a bit weird here, since the expression is not periondic in 2π in every direction as one would expect. I am prettys rue this comes from the fact that the distance between NN is the same as for the 0-flux case but the Brilouin zone got cut in hoalf in one direction.

represent hamiltonian as

$$H = \sum_{q,\sigma} \begin{pmatrix} a^\dagger_{q\sigma} & b^\dagger_{q\sigma} \end{pmatrix} \vec{h} \cdot \vec{\sigma} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \end{pmatrix} \tag{106}$$

$$\vec{h} = \begin{pmatrix} 2t\cos(k_y - k_x/2)) \\ 2t\sin(k_x/2) \\ 2t\cos(k_y) + \mu \end{pmatrix} \tag{107}$$

Then $\varepsilon(\vec{q})_\pm = \pm|\vec{h}|$ and

$$\frac{1}{\sqrt{2\varepsilon_+(\varepsilon_+ + 2t\cos(k_y) + \mu)}} \begin{pmatrix} \varepsilon_+ + 2t\cos(k_y) + \mu & -2t\cos(k_y - k_x/2) + 2ti\sin(k_x/2) \\ 2t\cos(k_y - k_x/2) + 2ti\sin(k_x/2) & \varepsilon_+ + 2t\cos(k_y) + \mu \end{pmatrix} \tag{108}$$

is the transformation matrix U. It has the properties that it is unitary and has determinant one. The expectation values are more involved bc of said matrix transformation. For example

$$\begin{aligned} \langle a^\dagger_{i\uparrow} b_{j\uparrow} \rangle &\overset{\text{FT}}{=} \frac{2}{N} \sum_{q,p} e^{i(\vec{q}\vec{R}_j - \vec{p}\vec{R}_i)} \underbrace{\langle a^\dagger_p b_q \rangle}_{\propto \delta_{q,p}} \\ &= \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} \langle a^\dagger_q b_q \rangle \\ &\overset{\text{BT}}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} \left(u^*_{11} u_{21} f(\varepsilon_+) + u^*_{12} u_{22} f(\varepsilon_-) \right) \end{aligned} \tag{109}$$

Simplify this using

1. $\varepsilon_+ = -\varepsilon_- \Rightarrow f(\varepsilon_+) + f(\varepsilon_-) = 1$
2. $u_{11} \in \mathbb{R}$
3. $u_{11} = u_{22}$
4. $u^*_{12} = -u_{21}$
5. $|u_{11}|^2 + |u_{12}|^2 = 1$ (normalization of eigenvectors)

$$\begin{aligned} u^*_{11} u_{21} f(\varepsilon_+) + u^*_{12} u_{22} f(\varepsilon_-) &\overset{1.}{=} (u^*_{11} u_{21} - u_{22} u^*_{12}) f(\varepsilon_+) + u_{22} u^*_{12} \\ &\overset{2.,3.,4.}{=} 2u_{11} u_{21} f(\varepsilon_+) - u_{11} u_{21} \\ &= u_{11} u_{21} (2f(\varepsilon_+) - 1) \end{aligned} \tag{110}$$

Due to the unit cell doubling the thermodynamic limit has to be slightly modified $\sum_k \to ((\Omega N)/2)/(2\pi)^2 \int dk$. (Each unit cell has 2 sites, this is why the factor 1/2 appears.)

$$\begin{aligned} \Rightarrow \langle a^\dagger_{i\uparrow} b_{j\uparrow} \rangle &= \Omega \int_{1.\text{BZ}} \frac{d^2q}{(2\pi)^2} e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} u_{11} u_{21} (2f(\varepsilon_+) - 1) \\ &= \int_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\left[i \left(k_y, 1/\sqrt{3}(k_x - k_y) \right)^T (\vec{R}_j - \vec{R}_i) \right] u_{11} u_{21} (2f(\varepsilon_+) - 1) \end{aligned} \tag{111}$$

$$u_{11} u_{21} = (t/\varepsilon)(\cos(k_y - k_x/2) + i\sin(k_x/2)) \tag{112}$$

Now for the other expectation values:

$$\Rightarrow \langle b^\dagger_{i\uparrow} a_{j\uparrow}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\left[i\left(k_y, 1/\sqrt{3}(k_x-k_y)\right)^T (\vec{R}_j - \vec{R}_i)\right] u_{11}u^*_{21}\,(2f(\varepsilon_+)-1) \tag{113}$$

$$\begin{aligned}
\langle a^\dagger_{i\uparrow} a_{j\uparrow}\rangle = \ldots &= \frac{2}{N}\sum_q e^{i\vec{q}(\vec{R}_j-\vec{R}_i)}\left[|u_{11}|^2 f(\varepsilon_+) + |u_{12}|^2 f(\varepsilon_-)\right] \\
&\overset{1.}{=} \frac{2}{N}\sum_q e^{i\vec{q}(\vec{R}_j-\vec{R}_i)}\left[|u_{12}|^2 - |u_{11}|^2) f(\varepsilon_-) + |u_{11}|^2\right] \\
&\overset{5.}{=} \frac{2}{N}\sum_q e^{i\vec{q}(\vec{R}_j-\vec{R}_i)}\left[(1-2u^2_{11}) f(\varepsilon_-) + u^2_{11}\right] \\
&\overset{1.}{=} \frac{2}{N}\sum_q e^{i\vec{q}(\vec{R}_j-\vec{R}_i)}\left[(1-2u^2_{11})(1-f(\varepsilon_+)) + u^2_{11}\right]
\end{aligned} \tag{114}$$

$$\Rightarrow \langle a^\dagger_{i\uparrow} a_{j\uparrow}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\left[i\left(k_y, 1/\sqrt{3}(k_x-k_y)\right)^T (\vec{R}_j - \vec{R}_i)\right] \left[(1-2u^2_{11})(1-f(\varepsilon_+)) + u^2_{11}\right] \tag{115}$$

$$\Rightarrow \langle b^\dagger_{i\uparrow} b_{j\uparrow}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\left[i\left(k_y, 1/\sqrt{3}(k_x-k_y)\right)^T (\vec{R}_j - \vec{R}_i)\right] \left[(1-2u^2_{11}) f(\varepsilon_+) + u^2_{11}\right] \tag{116}$$

$$u^2_{11} = (1/2\varepsilon_+)(\varepsilon_+ + 2t\cos(k_y) + \mu) \tag{117}$$

$$\langle N^a\rangle = 2\int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}(1-2u^2_{11}) f(\varepsilon_-) + u^2_{11} \tag{118}$$

$$\langle N^b\rangle = 2\int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}(1-2u^2_{11}) f(\varepsilon_+) + u^2_{11} \tag{119}$$

The constraint $\langle N^a_i\rangle = 1 = \langle N^b_i\rangle$ should, as gloal halffiling is already ensured and $\langle N^a\rangle$ and $\langle N^b\rangle$ are uniform by construction, be equivalent to $\langle N^a\rangle = \langle N^b\rangle$. Using the expressions above:

$$\langle N^a\rangle - \langle N^b\rangle = 2\int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}(1-2u^2_{11})(1-2f(\varepsilon_+)) \overset{!}{=} 0 \tag{120}$$

Numerics say that $\langle N^a\rangle = \langle N^b\rangle$ for $\mu = 0$ (and therefore $\mu_a = \mu_b = 0$). I feel like there is a more suphisticated argument to be made here, mabye even without the use of mathematica. Maybe one can find some symmetry argument using the form of $1-2f(\varepsilon_+)$.

Sanity check:

$$\langle N^a + N^b\rangle = 2\int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} 1 = 2 \tag{121}$$

Calculating the mean field parameters is less straight forward, as the formulae also depend on the types of sites they connect. The formulae are

$$\frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}a_{i+\delta_1\sigma}\rangle = \int\limits_{[-\pi,\pi]^2}\frac{d^2k}{(2\pi)^2}e^{ik_y}\left[(1-2u_{11}^2)(1-f(\varepsilon_+))+u_{11}^2\right] \tag{122}$$

$$\frac{1}{2}\sum_{\sigma}\langle b^{\dagger}_{i\sigma}b_{i+\delta_1\sigma}\rangle = \int\limits_{[-\pi,\pi]^2}\frac{d^2k}{(2\pi)^2}e^{ik_y}\left[(1-2u_{11}^2)f(\varepsilon_+)+u_{11}^2\right] \tag{123}$$

$$\frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}b_{i+\delta_3\sigma}\rangle = \int\limits_{[-\pi,\pi]^2}\frac{d^2k}{(2\pi)^2}e^{i(k_y-k_x/2)}u_{11}u_{21}(2f(\varepsilon_+)-1) \tag{124}$$

$$\frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}b_{i+\delta_2\sigma}\rangle = \int\limits_{[-\pi,\pi]^2}\frac{d^2k}{(2\pi)^2}e^{ik_x/2}u_{11}u_{21}^{*}(2f(\varepsilon_+)-1) \tag{125}$$

Consistency check: wehn reversing the direction of the bonds the $\vec{\delta}_2$ bond paramter should obtain a minus sign but the $\vec{\delta}_3$ bond parameter not. I wasnt able to figure this out by just looking at the formulae, so Ill just have to check by looking at the numerics for now.

Another observation:

$$\begin{aligned}\frac{1}{2}\sum_{\sigma}\left(\langle a^{\dagger}_{i\sigma}a_{i+\delta_1\sigma}\rangle + \langle b^{\dagger}_{i\sigma}b_{i+\delta_1\sigma}\rangle\right) &= \int\limits_{[-\pi,\pi]^2}\frac{d^2k}{(2\pi)^2}e^{ik_y}\\ &= \frac{1}{2\pi}\int\limits_{[-\pi,\pi]}dk_y e^{ik_y}\\ &= \frac{\sin(\pi)}{\pi}\\ &= 0\end{aligned} \tag{126}$$

$$\Rightarrow \frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}a_{i+\delta_1\sigma}\rangle = -\frac{1}{2}\sum_{\sigma}\langle b^{\dagger}_{i\sigma}b_{i+\delta_1\sigma}\rangle, \tag{127}$$

as expected. The numerical results are

$$\frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}a_{i+\delta_1\sigma}\rangle = 0.19754 \tag{128}$$

$$\frac{1}{2}\sum_{\sigma}\langle b^{\dagger}_{i\sigma}b_{i+\delta_1\sigma}\rangle = -0.19754 \tag{129}$$

$$\frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}b_{i+\delta_3\sigma}\rangle = 0.19754 \tag{130}$$

$$\frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}b_{i+\delta_2\sigma}\rangle = 0.19754 \tag{131}$$

$$\frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}b_{i-\delta_2\sigma}\rangle = -0.19754 \tag{132}$$

The corresponding hopping amplitudes are

$$t(\pm 0.19754) = \mp 0.87690J \tag{133}$$

$$\tag{134}$$

At last I will derive the energy density (since I had trouble with this before).

$$\begin{aligned}\frac{\langle H\rangle}{N} &= \frac{1}{N}\left\langle \sum_{q\sigma} \varepsilon_{+\sigma}\alpha^\dagger_{+\sigma}\alpha_{+\sigma} + \varepsilon_{-\sigma}\alpha^\dagger_{-\sigma}\alpha_{-\sigma}\right\rangle \\ &\overset{\text{th.lim.}}{\longrightarrow} 2\frac{1}{N}\frac{\Omega N}{2}\int\limits_{1.\text{BZ}} \frac{d^2q}{(2\pi)^2}\varepsilon_+ f(\varepsilon_+) + \varepsilon_- f(\varepsilon_-) \\ &= \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}\varepsilon_+(2f(\varepsilon_+)-1) \end{aligned} \tag{135}$$

(the blue 2 in the second line is the spin degeneracy)
Observation: the energy density is invariant under particle-hole transformation

$$\varepsilon_+(2f(\varepsilon_+)-1) \overset{\text{ph-trafo}}{\longrightarrow} \varepsilon_-(2f(\varepsilon_-)-1) = -\varepsilon_+(1-2f(\varepsilon_+)) = \varepsilon_+(2f(\varepsilon_+)-1) \tag{136}$$

The full energy density is:

$$\frac{\langle H\rangle}{N} = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}\varepsilon_+(\vec{k})(2f(\varepsilon_+(\vec{k}))-1) + 6\left(\tilde{J}\left(1-6\tilde{\kappa}|\langle\chi_{i,i+\delta}\rangle|^2\right)|\langle\chi_{i,i+\delta}\rangle|^2\right) = -1.2206\ J \tag{137}$$

4.3.2 Pedro's method

Here I apply Pedros method using the terms from my Hamiltonian. The free energy is

$$f_0 \equiv \frac{F_0(\kappa,\chi,T=0)}{N} = \frac{E_0}{N} + \frac{1}{N}\sum_{\vec{k}}\varepsilon(\vec{k}) \tag{138}$$

$$E_0 = 6N\left(\tilde{J}\left(1-6\tilde{\kappa}|\langle\chi_{ij}\rangle|^2\right)|\langle\chi_{ij}\rangle|^2\right) \tag{139}$$

$$\varepsilon(\vec{k}) = -2\left|-\tilde{J}\left(1-4\tilde{\kappa}|\langle\chi_{ij}\rangle|^2\right)\langle\chi_{ij}\rangle\right|\sqrt{\cos^2(k_1)+\cos^2(k_3)+\sin^2(k_2)} \tag{140}$$

From now on I use the shorthand $|\langle\chi_{ij}\rangle| \equiv \chi$.

$$\begin{aligned}\Rightarrow f_0 &= \tilde{J}\left(6(1-6\tilde{\kappa}\chi^2)\chi^2 - 2\gamma|\chi(1-4\tilde{\kappa}\chi^2)|\right) \\ &= \tilde{J}\left(\mp 2\gamma\chi + 6\chi^2 \pm 8\gamma\tilde{\kappa}\chi^3 - 36\tilde{\kappa}\chi^4\right)\end{aligned} \tag{141}$$

with

$$\gamma = \int\limits_{[-\pi,\pi]^2}\frac{d^2k}{(2\pi)^2}\sqrt{\cos^2(k_y)+\cos^2(k_y-k_x/2)+\sin^2(k_x/2)} \approx 1.18524 \tag{142}$$

Compared to Pedros results there is also a factor 2, besides other problems, because Pedro calculates f/M ($f/2$)!

$$\frac{\partial f_0}{\partial\chi} = \tilde{J}\left(\pm 2\gamma - 12\chi\right)\left(12\tilde{\kappa}\chi^2 - 1\right) \qquad \frac{\partial^2 f_0}{\partial\chi^2} = \tilde{J}\left(12 \pm 48\gamma\tilde{\kappa}\chi - 432\tilde{\kappa}\chi^2\right) \tag{143}$$

The extremal points of the free energy are at

$$\chi' = \gamma/6, \qquad \chi'' = \pm 1/\sqrt{12\tilde{\kappa}} \tag{144}$$

$-1/\sqrt{12\tilde{\kappa}}$ is always a Maximum. $\gamma/6$ is a minimum for $\kappa \leq 6/\gamma^2$, afterwards $1/\sqrt{12\tilde{\kappa}}$ becomes a minimum:

$$f_0 = \begin{cases} \frac{\tilde{J}\gamma^2}{6}\left(\frac{\tilde{\kappa}\gamma^2}{18} - 1\right), & \tilde{\kappa} \leq 6/\gamma^2 \\ \tilde{J}\left(\frac{1}{4\tilde{\kappa}} - \frac{2\gamma}{9}\sqrt{\frac{3}{\tilde{\kappa}}}\right), & \tilde{\kappa} > 6/\gamma^2 \end{cases} \tag{145}$$

$6/\gamma^2 \approx 2.1355$. For $\kappa = 10 \Rightarrow \tilde{\kappa} = 1.\bar{6}$ and therefore $\tilde{\kappa} < 6/\gamma^2$. Hence, the free energy minimum is at

$$\chi = \gamma/6 = 0.19754, \tag{146}$$

with the free energy

$$f_0 = -1.22206\, J. \tag{147}$$

These correspond to the values I calculated using the pedestrian method above. Mind that under particle-hole transformation the calculation doesn't change (the reason basically being the absolute value in the energy, or more physically the particle-hole symmetry of the system).

4.3.3 fermi surface

The dispersion relation reads

$$\varepsilon(\vec{q}) = 2|t|\sqrt{\cos^2(q_x) + \sin^2\left(\frac{1}{2}\left(q_x + \sqrt{3}q_y\right)\right) + \cos^2\left(\frac{1}{2}\left(q_x - \sqrt{3}q_y\right)\right)} \tag{148}$$

I will start by calculating the fermi surface $\varepsilon(\vec{q}) = 0$, which means that the expression under the square root must vanish. Using

$$\sin^2\left(\frac{1}{2}\left(q_x + \sqrt{3}q_y\right)\right) + \cos^2\left(\frac{1}{2}\left(q_x - \sqrt{3}q_y\right)\right) = 1 + \sin(q_x)\sin\left(\sqrt{3}q_y\right) \tag{149}$$

,
I obtain the following equation for the fermi surface:

$$\underbrace{1 + \cos^2(q_0^x)}_{\in[1,2]} = \underbrace{-\sin(q_0^x)\sin\left(\sqrt{3}q_0^y\right)}_{\in[-1,1]}. \tag{150}$$

The equation can be simplified by comparing the ranges of both sides (see above)

$$\Rightarrow 1 + \cos^2(q_0^x) \stackrel{!}{=} 1 \Rightarrow q_x^0 = \frac{\pi}{2} + \pi k, \quad k \in \mathbb{Z} \tag{151}$$

$$\Rightarrow \sin\left(\sqrt{3}q_0^y\right) = \begin{cases} -1 & \text{k even} \\ 1 & \text{k odd} \end{cases} \tag{152}$$

The solution (and thereby the Fermi surface) is

$$q_0^x = \frac{\pi}{2} + \pi k \tag{153}$$

$$q_0^y = \frac{1}{\sqrt{3}}\left(\frac{(-1)^{k+1}\pi}{2} + 2\pi m\right), \tag{154}$$

where $k, m \in \mathbb{Z}$. This means that the fermi surface is a collection of isolated points. More scientifically spoken: the state is a semi-metal. Two of those Fermi-points lie in the first Brillouin zone:

$$\vec{K} = \frac{\pi}{2}\begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix} \quad \text{(k = m =0)} \tag{155}$$

$$\vec{K}' = \frac{\pi}{2}\begin{pmatrix} -1 \\ 1/\sqrt{3} \end{pmatrix} \quad \text{(k = -1, m =0).} \tag{156}$$

Mind you that these points are not equivalent! Evidently they are time-reversal partners.

4.3.4 small momentum expansion

To access the low-energy physics we must approximate the Hamiltonian around the fermi-points using

$$f\left(\vec{q}_0 + \delta\vec{q}\right) \approx f(\vec{q}_0) + \left(\nabla_{\vec{q}} f(\vec{q})|_{\vec{q}=\vec{q}_0}\right) \cdot \delta\vec{q}. \tag{157}$$

Around $\vec{K}$:

$$\cos\left(\vec{\delta}_1(\vec{K}+\vec{p})\right) \approx \cos\left(\vec{\delta}_1\vec{K}\right) - \sin\left(\vec{\delta}_1\vec{K}\right)\vec{\delta}_1\vec{p} = -\vec{\delta}_1\vec{p} \tag{158}$$

$$\sin\left(\vec{\delta}_2(\vec{K}+\vec{p})\right) \approx \sin\left(\vec{\delta}_2\vec{K}\right) + \cos\left(\vec{\delta}_2\vec{K}\right)\vec{\delta}_2\vec{p} = \vec{\delta}_2\vec{p} \tag{159}$$

$$\cos\left(\vec{\delta}_3(\vec{K}+\vec{p})\right) \approx \cos\left(\vec{\delta}_3\vec{K}\right) - \sin\left(\vec{\delta}_3\vec{K}\right)\vec{\delta}_3\vec{p} = -\vec{\delta}_3\vec{p} \tag{160}$$

Use

$$\vec{\delta}_1\vec{K} = \frac{\pi}{2} \tag{161}$$

$$\vec{\delta}_2\vec{K} = 0 \tag{162}$$

$$\vec{\delta}_3\vec{K} = \frac{\pi}{2} \tag{163}$$

to obtain

$$\Rightarrow H_{\vec{K}} \approx 2t\sum_{p,\sigma}\begin{pmatrix} a^\dagger_{p\sigma} & b^\dagger_{p\sigma} \end{pmatrix}\begin{pmatrix} -\vec{\delta}_1\vec{p} & \left(-\vec{\delta}_3 - i\vec{\delta}_2\right)\vec{p} \\ \left(-\vec{\delta}_3 + i\vec{\delta}_2\right)\vec{p} & \vec{\delta}_1\vec{p} \end{pmatrix}\begin{pmatrix} a_{p\sigma} \\ b_{p\sigma} \end{pmatrix} \tag{164}$$

And around $\vec{K}' = -\vec{K}$:

$$\Rightarrow H_{\vec{K}'} \approx 2t\sum_{p,\sigma}\begin{pmatrix} a^\dagger_{p\sigma} & b^\dagger_{p\sigma} \end{pmatrix}\begin{pmatrix} \vec{\delta}_1\vec{p} & \left(\vec{\delta}_3 - i\vec{\delta}_2\right)\vec{p} \\ \left(\vec{\delta}_3 + i\vec{\delta}_2\right)\vec{p} & -\vec{\delta}_1\vec{p} \end{pmatrix}\begin{pmatrix} a_{p\sigma} \\ b_{p\sigma} \end{pmatrix} \tag{165}$$

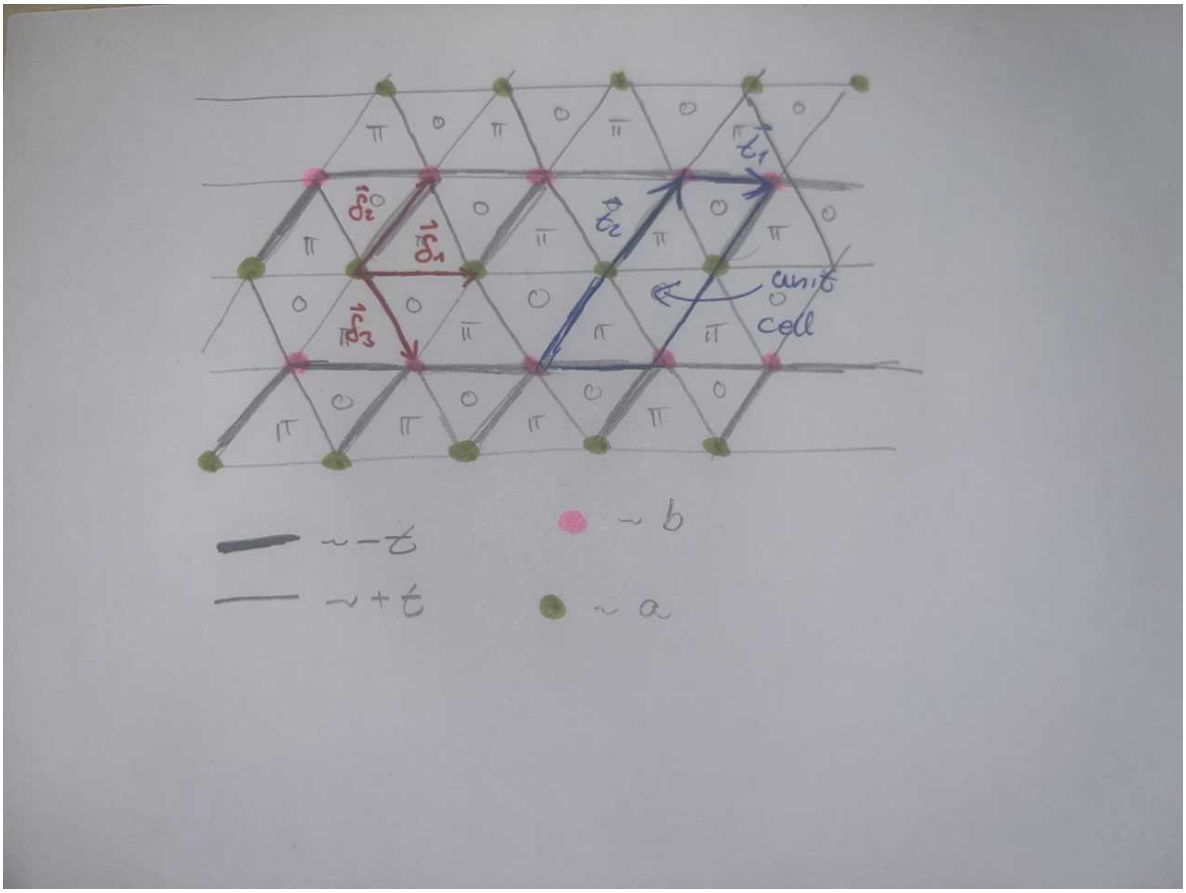
This gives the dispersion relation (for both cones)

$$\begin{aligned} \varepsilon &\approx 2t\sqrt{p_x^2 + \frac{1}{4}\left(p_x + \sqrt{3}p_y\right)^2 + \frac{1}{4}\left(p_x - \sqrt{3}p_y\right)^2} \\ &= \sqrt{6}t\sqrt{p_x^2 + p_y^2} \\ &= \sqrt{6}t|\vec{p}| \end{aligned} \tag{166}$$

Thus, the fermi velocity is $v_F = \sqrt{6}t$. This result has been reported before here [**?**]. $\vec{K}$ and $\vec{K}'$ are time revearsel partners, thus the two hamiltonains should map onto each other under time revearesal. I would guess that $T = -C$ in k-space, where C is complex conjugation, which would be consistent with the form of the Hamiltonian.

4.3.5 Continuum Model/ Gradient Expansion

Next I will derive the low-energy Hamilotnian in real space, as a continuum theory (/field theory). I start by writing down the real-space hamiltonian, I reshuffled term a bit compared to when I first wrote it down above, the chemical potentials are set to zero.



$$H = \frac{t}{2}\sum_{i\sigma}\left(a^\dagger_{i\sigma}a_{i+\delta_1;\sigma} - a^\dagger_{i\sigma}b_{i+\delta_2;\sigma} + a^\dagger_{i\sigma}b_{i+\delta_3;\sigma} - b^\dagger_{i\sigma}b_{i+\delta_1;\sigma} + b^\dagger_{i\sigma}a_{i+\delta_2;\sigma} + b^\dagger_{i\sigma}a_{i+\delta_3;\sigma} + \text{h.c.}\right) \tag{167}$$

I chose a bad method for indexing the lattice position, I should eventually revise this.

The idea is to decompose the the real-space wavefuntion by its momentum - one part is the momentum is the momentum of the dirac point, the other point describes a small deviation from the dirac point (small wavevector -¿ large wavelength -¿ slowly varying in real space). This justifies a gradient expansin of the "deviation field". think of a better explenation for this. But I guess its correct in its spirit.

$$\begin{aligned} a_{i\sigma} &\sim e^{i\vec{K}\vec{R}_i}A_{K\sigma}(\vec{R}_i) + e^{i\vec{K}'\vec{R}_i}A_{K'\sigma}(\vec{R}_i) \\ b_i &\sim e^{i\vec{K}\vec{R}_i}B_{K\sigma}(\vec{R}_i) + e^{i\vec{K}'\vec{R}_i}B_{K'\sigma}(\vec{R}_i) \end{aligned} \tag{168}$$

$A_{K\sigma}, A_{K'\sigma}, B_{K\sigma}$ aqnd $B_{K'\sigma}$ are slowly varying fermionic fields. I begin with the expansion around the K-point. If the above expansion is correct, the cross terms should vanish. Show this. The protoypical method is

$$
\begin{aligned}
a^\dagger_{i\sigma} a_{i+\delta;\sigma} &\sim e^{i\vec{K}\vec{\delta}} A^\dagger_{K\sigma}(\vec{R}_i) A_{K\sigma}(\vec{R}_i + \vec{\delta}) \\
&\approx e^{i\vec{K}\vec{\delta}} A^\dagger_{K\sigma}(\vec{R}_i) \left(1 + \vec{\delta}\vec{\nabla}_{\vec{R}}\right) A_{K\sigma}(\vec{R}_i) \\
&= e^{i\vec{K}\vec{\delta}} A^\dagger_{K\sigma}(\vec{R}_i) \left(1 + i\vec{\delta}\vec{p}\right) A_{k\sigma}(\vec{R}_i),
\end{aligned} \tag{169}
$$

where $\vec{p} = -i\vec{\nabla}_{\vec{R}}$. The diagonal terms are

$$
A^\dagger_{K\sigma}(\vec{R}_i) \left[e^{i\vec{K}\vec{\delta}_1}(1 + i\vec{\delta}_1\vec{p}) + e^{-i\vec{K}\vec{\delta}_1}(1 - i\vec{\delta}_1\vec{p})\right] A_{K\sigma}(\vec{R}_i) \tag{170}
$$

$$
B^\dagger_{K\sigma}(\vec{R}_i) \left[-e^{i\vec{K}\vec{\delta}_1}(1 + i\vec{\delta}_1\vec{p}) - e^{-i\vec{K}\vec{\delta}_1}(1 - i\vec{\delta}_1\vec{p})\right] B_{K\sigma}(\vec{R}_i). \tag{171}
$$

The off-diagonal terms are

$$
A^\dagger_{K\sigma}(\vec{R}_i) \frac{t}{2} \left[-e^{i\vec{\delta}_2\vec{K}}(1 + i\vec{\delta}_2\vec{p}) + e^{-i\vec{\delta}_2\vec{K}}(1 - i\vec{\delta}_2\vec{p}) + e^{i\vec{\delta}_3\vec{K}}(1 + i\vec{\delta}_3\vec{p}) + e^{-i\vec{\delta}_3\vec{K}}(1 - i\vec{\delta}_3\vec{p})\right] B_{K\sigma}(\vec{R}_i) \tag{172}
$$

$$
B^\dagger_{K\sigma}(\vec{R}_i) \frac{t}{2} \left[e^{i\vec{\delta}_2\vec{K}}(1 + i\vec{\delta}_2\vec{p}) - e^{-i\vec{\delta}_2\vec{K}}(1 - i\vec{\delta}_2\vec{p}) + e^{i\vec{\delta}_3\vec{K}}(1 + i\vec{\delta}_3\vec{p}) + e^{-i\vec{\delta}_3\vec{K}}(1 - i\vec{\delta}_3\vec{p})\right] A_{K\sigma}(\vec{R}_i) \tag{173}
$$

Use

$$
\vec{\delta}_1\vec{K} = \frac{\pi}{2} \tag{174}
$$

$$
\vec{\delta}_2\vec{K} = 0 \tag{175}
$$

$$
\vec{\delta}_3\vec{K} = \frac{\pi}{2} \tag{176}
$$

to obtain

$$
H_{\vec{K}} \approx \sum_\sigma 2t \int d^2\vec{R} \quad \Psi^\dagger_{K\sigma}(\vec{R}) \begin{pmatrix} -\vec{\delta}_1\vec{p} & \left(-\vec{\delta}_3 - i\vec{\delta}_2\right)\vec{p} \\ \left(-\vec{\delta}_3 + i\vec{\delta}_2\right)\vec{p} & \vec{\delta}_1\vec{p} \end{pmatrix} \Psi_{K\sigma}(\vec{R}), \tag{177}
$$

with

$$
\Psi_{K\sigma}(\vec{R}) = \begin{pmatrix} A_{K\sigma}(\vec{R}) \\ B_{K\sigma}(\vec{R}) \end{pmatrix}. \tag{178}
$$

Doing the same around $\vec{K}'$ gives

$$
H_{\vec{K}'} \approx \sum_\sigma 2t \int d^2\vec{R} \quad \Psi^\dagger_{K'\sigma}(\vec{R}) \begin{pmatrix} \vec{\delta}_1\vec{p} & \left(\vec{\delta}_3 - i\vec{\delta}_2\right)\vec{p} \\ \left(\vec{\delta}_3 + i\vec{\delta}_2\right)\vec{p} & -\vec{\delta}_1\vec{p} \end{pmatrix} \Psi_{K'\sigma}(\vec{R}) \tag{179}
$$

Maybe it is handy to write $\vec{\delta}_1 = \vec{\delta}_2 + \vec{\delta}_3$. When Comparing this to the results obtained by expanding in reciprocal space, keep in mind that here $\vec{p}$ is an operator. So technically, one would have to do a kind of fourier transform for the fermionic fields to find the exact representation as found in recriprocal space. This is why the Hamiltonian looks so similar, even though we are still in real space here.

Imagine we did this small frequency? Fourier expansion and we found the reciprocal space Hamiltonian (replace the momentum operator by its eigenvalue!), the corresponding disperion relation would then read

$$\varepsilon(\vec{p}) = \left| 2t \begin{pmatrix} \mp\vec{\delta}_3\vec{p} \\ \vec{\delta}_2\vec{p} \\ \mp\vec{\delta}_1\vec{p} \end{pmatrix} \right| = \sqrt{6}t\sqrt{p_x^2 + p_y^2} = \sqrt{6}t|\vec{p}|, \tag{180}$$

which complies with the k-space result. In case I want to diagonalize this in operator form at some point; since the problem is non-interacting there must be a harmonic oscillator hidden in there.

4.4 0-π State on Strained Lattice

references are [?], [?], [?], [?] and [?]

4.4.1 Basics of Strain

For the purpose of introducing strain into the model, I first establish its effect on the coordinates and exchange couplings

$$\vec{R}_i \mapsto \vec{R}'_i = \vec{R}_i + \vec{U}(\vec{R}_i) \tag{181}$$
$$\vec{\delta}_{ij} \mapsto \vec{\delta}'_{ij} = \vec{\delta}_{ij} + \vec{U}(\vec{R}_j) - \vec{U}(\vec{R}_i) \tag{182}$$

$\vec{\delta}_{ij} = \vec{R}_j - \vec{R}_i$;)

$\vec{U}(\vec{R})$ is the displacement field. We approximate that $\vec{U}$ varies slowly with $\vec{R}$:

$$\vec{U}(\vec{R}_j) = \vec{U}(\vec{R}_i + \vec{\delta}_{ij}) \approx \vec{U}(\vec{R}_i) + \left(\vec{\delta}_{ij}\vec{\nabla}_{\vec{R}}\right)\vec{U}(\vec{R})\big|_{\vec{R}_i} \tag{183}$$
$$\tag{184}$$

It is convenient to redefine

$$\left(\vec{\delta}_{ij}\vec{\nabla}_{\vec{R}}\right)\vec{U}(\vec{R})\big|_{\vec{R}_i} = \underline{u}\cdot\vec{\delta}_{ij}, \tag{185}$$

where $u_{\alpha\beta} = \partial_\beta U_\alpha$ is the strain tensor.

$$\Rightarrow \vec{\delta}'_{ij} \approx \vec{\delta}_{ij} + \underline{u}\cdot\vec{\delta}_{ij} \tag{186}$$

Here $\underline{u}$ is taken at $\vec{R}_i$, however one can derive the same expression but with $\underline{u}$ taken at $\vec{R}_j$. In what follows I choose the expansion around $\vec{R}_i$ as shown in the main text. This however is arbitrary. Something dependening on $\vec{R}_i$ and $\vec{R}_j$ (or $(\vec{R}_i + \vec{R}_j)/2$) would seem more sensible but would complicate later expressions. I was not able to resolve this yet.

I do the same approximation for the exchange couplings

$$J(\vec{\delta}_{ij}) \mapsto J(\vec{\delta}'_{ij}) \approx J\left(\vec{\delta}_{ij} + \underline{u}\cdot\vec{\delta}_{ij}\right) \tag{187}$$
$$\approx J(\vec{\delta}_{ij}) + \left(\underline{u}\cdot\vec{\delta}_{ij}\right)\cdot\left(\vec{\nabla}_{\vec{\delta}}J(\vec{\delta})\big|_{\delta_{ij}}\right) \tag{188}$$

For the function $J(\vec{\delta})$ I uses the form

$$J(\vec{\delta}) = J\exp\left[-\beta\left(|\vec{\delta}|/a - 1\right)\right] \approx J\left(1 - \beta\left(|\vec{\delta}|/a - 1\right)\right), \tag{189}$$

, see e.g. [?], where a is the lattice constant and $\beta = -d\ln(J)/d\ln\left(|\vec{\delta}|\right)\big|_a$ the magneto-elastic coupling.

$$\Rightarrow \vec{\nabla}_{\vec{\delta}} J(\vec{\delta})\big|_{\delta_{ij}} \overset{|\vec{\delta}_{ij}|=a}{=} -\frac{J\beta}{a^2}\vec{\delta}_{ij}$$
$$\Rightarrow J(\vec{\delta}_{ij}) \approx J(\vec{\delta}_{ij}) - \frac{J\beta}{a^2}\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \tag{190}$$

Looking at the matrix elments (170 and below), one also requires the expansion of $\exp\left[i\vec{\delta}_{ij}\vec{K}\right]$. To summarize:

$$\vec{\delta}'_{ij} \approx \vec{\delta}_{ij} + \underline{u}\cdot\vec{\delta}_{ij} \tag{191}$$
$$J(\vec{\delta}'_{ij}) \approx J - \frac{J\beta}{a^2}\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \tag{192}$$
$$e^{i\vec{K}\cdot\vec{\delta}'_{ij}} \approx e^{i\vec{K}\cdot\vec{\delta}_{ij}} + ie^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \tag{193}$$

In this approximation the Mean field does not react to the strain at all??

4.4.2 Straining the 0-π-State

The prototypical form of the matrix elements is (expanding around $\vec{K}$)

$$J(\vec{\delta}_{ij})a^\dagger_{i\sigma}a_{i+\delta_{ij};\sigma} \sim A^\dagger_{K\sigma}\left(J(\vec{\delta}_{ij})e^{i\vec{K}\cdot\vec{\delta}_{ij}}(1+i\vec{\delta}_{ij}\cdot\vec{p})\right)A_{K\sigma} \tag{194}$$

Expanding up to first order in strain I acquire the following terms

$Je^{i\vec{K}\cdot\vec{\delta}_{ij}}(1+i\vec{\delta}_{ij}\cdot\vec{p})$	unstrained Hamiltonian/ zeroth order
$iJe^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)$	contribution to $\vec{A}$ of $\mathcal{O}\left(\underline{u}^1\right)$
$-\frac{J\beta}{a^2}e^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)$	contribution to $\vec{A}$ of $\mathcal{O}\left(\underline{u}^1\right)$
$-Je^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)\left(\vec{\delta}_{ij}\cdot\vec{p}\right)$	correction to v_F of $\mathcal{O}\left(\underline{u}^1\right)$
$-i\frac{J\beta}{a^2}e^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)\left(\vec{\delta}_{ij}\cdot\vec{p}\right)$	correction to v_F of $\mathcal{O}\left(\underline{u}^1\right)$
$iJe^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(\underline{u}\cdot\vec{\delta}_{ij}\right)\cdot\vec{p}$	correction to v_F of $\mathcal{O}\left(\underline{u}^1\right)$

(I drop the lattice constant again from now on) I will focus on the terms of order zero + terms of either order 1 in field gradient or order 1 in strain [**?**]. I havent come up with a good argument for this yet. How does one even compares orders between different expansions? This gives

$$\begin{aligned} J(\vec{\delta}_{ij})e^{i\vec{K}\cdot\vec{\delta}_{ij}}(1+i\vec{\delta}_{ij}\cdot\vec{p}) &\sim Je^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(1+i\vec{\delta}_{ij}\cdot\vec{p}+i\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)-\beta\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)\right) \\ &= Je^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(1+i\vec{\delta}_{ij}\cdot\left[\vec{p}+\underline{u}^T\left(\vec{K}+i\beta\vec{\delta}_{ij}\right)\right]\right) \end{aligned} \tag{195}$$

Hence, on this level

$$\vec{A} = \underline{u}^T\left(\vec{K}+i\beta\vec{\delta}_{ij}\right). \tag{196}$$

This is not correct since one in has to add contributions from different local directions in the off-diagonal terms of H!

$$B = (\partial_x A_y - \partial_y A_x) \tag{197}$$

It turns out however, that $\vec{B} = 0$, which is explained in what follows. The first term is simply rotationless (this was shown here [**?**]) and can therefore be "gauged away".

Regarding the second term; I take a step back to the dispersion relation of the unstrained model

$$H = \sum_{\vec{q}\sigma} \Psi^{\dagger}_{q\sigma} \cdot \left(\vec{h}(\vec{q}) \cdot \vec{\sigma}\right) \cdot \Psi_{q\sigma} \quad \text{with} \quad \vec{h}(\vec{q}) = 2t \begin{pmatrix} \sin\left(\vec{q}\cdot\vec{\delta}_3\right) \\ \cos\left(\vec{q}\cdot\vec{\delta}_2\right) \\ \cos\left(\vec{q}\cdot\vec{\delta}_1\right) \end{pmatrix} \Rightarrow \varepsilon(\vec{q}) = \left|\vec{h}(\vec{q})\right|. \tag{198}$$

By definition, $\varepsilon(\vec{K}^{(\prime)}) = 0$. For this to be true, every component of $\vec{h}$ must vanish individually at $\vec{K}^{(\prime)}$. This gives three conditions for $\vec{q}$, one for each direction.

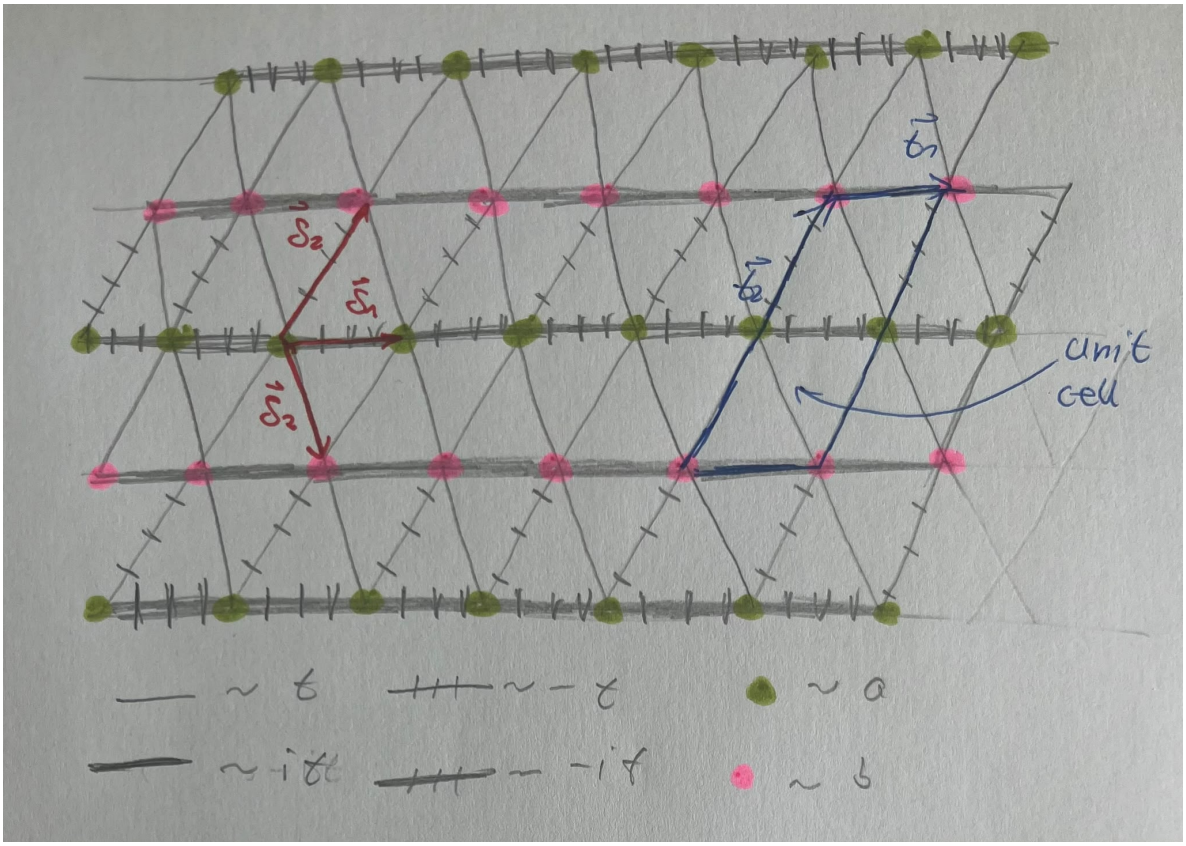
Evidently each expression in $\vec{h}$ is obtained by adding contribtions from hopping forwards and backwards along $\vec{\delta}_{ij}$. However, since $\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij}$ is invariant under $\vec{\delta}_{ij} \mapsto -\vec{\delta}_{ij}$ it acquires the same dispersion-like prefactor as the zeroth order, which vanishes by definition at $\vec{K}^{(\prime)}$.

So why does the term vanish? Its a mix of the special structure of the hamiltonian (i.e. the local directions decouple in the pauli-matrix basis) and the fact that the expansion term $\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij}$ cant distinguish between $\vec{\delta}_{ij}$ and $-\vec{\delta}_{ij}$. The latter statement is obviously not model-specific. The same mechanism shouild be present in the π-flux square lattice. For uniform hopping (e.g. ordinary Graphene) it is absent (the decoupling of the directions is a cosnequence of the non-uniform hopping/ the mean-field).

It is not clear to me how this can be upheld under gauge transformations (wouldn't this mix directions in the Pauli-matrix basis?)

I would expect to find a pseudo-magnetic field in higher orders of strain?

4.5 $\pi/2$-$\pi/2$ and $-\pi/2$ - $-\pi/2$ States



4.5.1 Pedestrian Method

This is very similar to the $\pi - 0-$flux-state. The bare Hamiltonian is

$$\begin{aligned} H = \sum_{i,\sigma} \Bigg(\frac{t}{2} \Big[& - i a^\dagger_{i\sigma} a_{i+\delta_1;\sigma} + i a^\dagger_{i\sigma} a_{i-\delta_1;\sigma} - a^\dagger_{i\sigma} b_{i+\delta_2;\sigma} + a^\dagger_{i\sigma} b_{i-\delta_2;\sigma} + a^\dagger_{i\sigma} b_{i+\delta_3;\sigma} + a^\dagger_{i\sigma} b_{i-\delta_3;\sigma} \\ & + i b^\dagger_{i\sigma} b_{i+\delta_1;\sigma} - i b^\dagger_{i\sigma} b_{i-\delta_1;\sigma} + b^\dagger_{i\sigma} a_{i+\delta_2;\sigma} - b^\dagger_{i\sigma} a_{i-\delta_2;\sigma} + a^\dagger_{i\sigma} b_{i+\delta_3;\sigma} + a^\dagger_{i\sigma} b_{i-\delta_3;\sigma} \Big] \\ & + \mu_a a^\dagger_{i\sigma} a_{i\sigma} + \mu_b b^\dagger_{i\sigma} b_{i\sigma} \Bigg) - \frac{N(\mu_a + \mu_b)}{2} \end{aligned} \tag{199}$$

The site index notation is not ideal, as a and b sites by definition cant have the same site indices.

$$\Rightarrow H = \sum_{q,\sigma} \begin{pmatrix} a^\dagger_{q\sigma} & b^\dagger_{q\sigma} \end{pmatrix} \begin{pmatrix} 2t \sin\left(\vec{q}\vec{\delta}_1\right) + \mu_a & 2t \cos\left(\vec{q}\vec{\delta}_3\right) - 2it \sin\left(\vec{q}\vec{\delta}_2\right) \\ 2t \cos\left(\vec{q}\vec{\delta}_3\right) + 2it \sin\left(\vec{q}\vec{\delta}_2\right) & -2t \sin\left(\vec{q}\vec{\delta}_1\right) + \mu_b \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \end{pmatrix}. \tag{200}$$

Maybe I should show explicitly that $\mu = 0$, I guess $\mu_a = -\mu_b = \mu$ is clear
Using $\mu_a = \mu_b = 0$:

$$\vec{h} = 2t \begin{pmatrix} \cos\left(\vec{q}\vec{\delta}_1\right) \\ \sin\left(\vec{q}\vec{\delta}_2\right) \\ \sin\left(\vec{q}\vec{\delta}_3\right) \end{pmatrix} \mapsto 2t \begin{pmatrix} \cos(k_y - k_x/2) \\ \sin(k_x/2) \\ \sin(k_y) \end{pmatrix} \tag{201}$$

$$\varepsilon(\vec{k})_{\pm} = \pm 2|t|\sqrt{\sin^2(k_y) + \cos^2(k_y - k_x/2) + \sin^2(k_x/2)} \tag{202}$$

$$\frac{1}{\sqrt{2\varepsilon_+(\varepsilon_+ + 2t\sin(k_y))}} \begin{pmatrix} \varepsilon_+ + 2t\sin(k_y) & -2t\cos(k_y - k_x/2) + 2ti\sin(k_x/2) \\ 2t\cos(k_y - k_x/2) + 2ti\sin(k_x/2) & \varepsilon_+ + 2t\sin(k_y) \end{pmatrix} \tag{203}$$

$$u_{11}u_{21} = (t/\varepsilon_+)(\cos(k_y - k_x/2) + i\sin(k_x/2)) \tag{204}$$

$$u_{11}^2 = (1/2\varepsilon_+)(\varepsilon_+ + 2t\sin(k_y)) \tag{205}$$

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger a_{i+\delta_1\sigma}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_y}\left[(1-2u_{11}^2)(1-f(\varepsilon_+)) + u_{11}^2\right] \tag{206}$$

$$\frac{1}{2}\sum_\sigma \langle b_{i\sigma}^\dagger b_{i+\delta_1\sigma}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_y}\left[(1-2u_{11}^2)f(\varepsilon_+) + u_{11}^2\right] \tag{207}$$

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger b_{i+\delta_3\sigma}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{i(k_y - k_x/2)}u_{11}u_{21}(2f(\varepsilon_+) - 1) \tag{208}$$

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger b_{i+\delta_2\sigma}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_x/2}u_{11}u_{21}^*(2f(\varepsilon_+) - 1) \tag{209}$$

One has to be a bit careful of where the sign of t enters when choosing $t = 1$ for the self consistency. Due to the complex conjugation the σ_z-terms are insensitive to the sign of t, but the (real) σ_x- and σ_y-terms are not. The results (choosing $t = 1$ are):

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger a_{i+\delta_1\sigma}\rangle = -0.200169i \tag{210}$$

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger a_{i-\delta_1\sigma}\rangle = 0.200169i \tag{211}$$

$$\frac{1}{2}\sum_\sigma \langle b_{i\sigma}^\dagger b_{i+\delta_1\sigma}\rangle = 0.200169i \tag{212}$$

$$\frac{1}{2}\sum_\sigma \langle b_{i\sigma}^\dagger b_{i-\delta_1\sigma}\rangle = -0.200169i \tag{213}$$

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger b_{i+\delta_3\sigma}\rangle = 0.200169 \tag{214}$$

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger b_{i-\delta_3\sigma}\rangle = 0.200169 \tag{215}$$

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger b_{i+\delta_2\sigma}\rangle = 0.200169 \tag{216}$$

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger b_{i-\delta_2\sigma}\rangle = -0.200169 \tag{217}$$

This complies with the cartoon on top. The value for t is

$$|t_{ij}| = \left| J\langle\chi_{ij}\rangle^*(1+\kappa/2-4\kappa|\langle\chi_{ij}\rangle|^2 \right| \overset{\kappa=10}{=} 0.88020\, J. \tag{218}$$

The free energy is

$$\frac{\langle H\rangle}{N} = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}\varepsilon_+(\vec{k})(2f(\varepsilon_+(\vec{k}))-1) + 6\left(\tilde{J}\left(1-6\tilde{\kappa}|\langle\chi_{i,i+\delta}\rangle|^2\right)|\langle\chi_{i,i+\delta}\rangle|^2\right) = -1.24978\ J \tag{219}$$

4.5.2 Pedro's Method

The expression for the free energy is identical to the $0-\pi$-state except that γ gets replaced by a different constant α:

$$\alpha = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}\sqrt{\sin^2(k_y)+\cos^2(k_y-k_x/2)+\sin^2(k_x/2)} \approx 1.20102 \tag{220}$$

Crucally α is larger than gamma. The results for this method are

$$|\chi| = \alpha/6 = 0.20017, \tag{221}$$

$$f_0 = \tilde{J}\left(\frac{1}{4\tilde{\kappa}} - \frac{2\alpha}{9}\sqrt{\frac{3}{\tilde{\kappa}}}\right) = -1.24979\, J \tag{222}$$

The methods agree.

4.5.3 Chern number

Here I will calculate the Chern number associated with the state. The relevant formula for a 2-band system is is

$$C = \frac{1}{4\pi}\int\limits_{1.\mathrm{BZ}} d^2q\,\hat{h}\cdot\left(\partial_{q_x}\hat{h}\times\partial_{q_y}\hat{h}\right), \tag{223}$$

where $\hat{h}$ is the normalized Bloch vector (is this the correct terminology?).

$$\vec{h} = 2t\begin{pmatrix}\cos\left(\frac{1}{2}(q_x-\sqrt{3}q_y)\right)\\ \sin\left(\frac{1}{2}(q_x+\sqrt{3}q_y)\right)\\ \sin(q_x)\end{pmatrix} \equiv 2t\begin{pmatrix}\cos(\delta_3)\\ \sin(\delta_2)\\ \sin(\delta_1)\end{pmatrix}, \quad \left|\vec{h}\right| \equiv \varepsilon(\vec{q}) \tag{224}$$

I first simplify 223 a bit using

$$\partial_\alpha\hat{h} = \frac{1}{\left|\vec{h}\right|}\left(\partial_\alpha\vec{h} - \left(\hat{h}\cdot\partial_\alpha\vec{h}\right)\hat{h}\right). \tag{225}$$

When applying this to 223, 3 out of the for terms that appear are proportional to $\hat{h}\times\hat{h}=\vec{0}$.

$$\Rightarrow C = \frac{1}{4\pi}\int\limits_{1.BZ} d^2q\,[\varepsilon(\vec{q})]^{-3}\vec{h}\cdot\left(\partial_{q_x}\vec{h}\times\partial_{q_y}\vec{h}\right) \tag{226}$$

$$\partial_{q_x}\vec{h} = t\begin{pmatrix}-\sin(\delta_3)\\ \cos(\delta_2)\\ 2\cos(\delta_1)\end{pmatrix}, \qquad \partial_{q_y}\vec{h} = \sqrt{3}t\begin{pmatrix}\sin(\delta_1)\\ \cos(\delta_2)\\ 0\end{pmatrix} \tag{227}$$

$$\Rightarrow \vec{h}\cdot\left(\partial_{q_x}\vec{h}\times\partial_{q_y}\vec{h}\right) = 4\sqrt{3}t^3\left(-\cos(\delta_1)\cos(\delta_2)\cos(\delta_3)+\cos(\delta_1)\sin(\delta_2)\sin(\delta_3)-\sin(\delta_1)\cos(\delta_2)\sin(\delta_3)\right)$$
$$= 4\sqrt{3}t^3\left(\sin(\delta_1)\sin(\delta_2)\cos(\delta_3)-1\right) \tag{228}$$

$$\Rightarrow C = \frac{\sqrt{3}}{8\pi}\int\limits_{1.\mathrm{BZ}} d^2q \frac{\sin(\delta_1)\sin(\delta_2)\cos(\delta_3)-1}{\left(\sin^2(\delta_1)+\sin^2(\delta_2)+\cos^2(\delta_3)\right)^{3/2}} \tag{229}$$

For evaluation I use the transformation:

$$\begin{pmatrix} q_x \\ q_y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1\sqrt{3} & -1/\sqrt{3} \end{pmatrix}\begin{pmatrix} k_x \\ k_y \end{pmatrix} \tag{230}$$

$$\Rightarrow C = \frac{1}{8\pi}\int\limits_{[-\pi,\pi]^2} d^2k \frac{\sin(k_y)\sin(k_y-k_x/2)\cos(k_x/2)-1}{\left(\sin^2(k_y)+\sin^2(k_y-k_x/2)+\cos^2(k_x/2)\right)^{3/2}} \tag{231}$$

The result is $C=-1$.

5. Kagome Lattice

5.1 Precursor: Methods

5.1.1 (smartly) projecting onto a 2-Band Subspace

In what follows we will encounter systems with more than two bands. In order to derive an effective theory for a small energy window around a Dirac-point most of these bands are irrelevant and can be safely projected out. Asssume a k-space Hamiltonian $H(\vec{q})$ and a degeneracy point (/Dirac point) $\vec{K}$. Furtheremore, I define

$$H(\vec{K}+\vec{p}) \approx \underbrace{H(\vec{K})}_{\equiv H^0_{\vec{K}}} + \underbrace{\vec{p}\cdot\left(\vec{\nabla}_{\vec{q}}H(\vec{q})\right)\Big|_{\vec{q}=\vec{K}}}_{\equiv H^1_{\vec{K}}} + \mathcal{O}(p^2). \tag{232}$$

The next ingredient is a orthonormal basis of the degenerate subspace $\{\vec{v}_1,\vec{v}_2\}$. I also need the matrix elements of $H^1_{\vec{K}}$ wrt to this basis:

$$\begin{aligned}
\vec{v}_1^H\cdot H^1_{\vec{K}}\cdot\vec{v}_1 &\equiv M_{11} = M^x_{11}p_x + M^y_{11}p_y \\
\vec{v}_1^H\cdot H^1_{\vec{K}}\cdot\vec{v}_2 &\equiv M_{12} = M^x_{12}p_y + M^y_{12}p_y \\
\vec{v}_2^H\cdot H^1_{\vec{K}}\cdot\vec{v}_2 &= -M_{11} \\
\vec{v}_2^H\cdot H^1_{\vec{K}}\cdot\vec{v}_2 &= (M_{12})^*
\end{aligned} \tag{233}$$

The last two identities are motivated by the effect that $(M)_{ij}=M_{ij}$ must be a Hermitian matrix with no absolute part ($\propto\sigma_0$). To obtain an arbitrary basis of the degnerate subspace I do a $SU(2)$-rotation ($SU(2)$ automatically enforces the orthonormality of the new basis):

$$\begin{pmatrix} \vec{w}_1 \\ \vec{w}_2 \end{pmatrix} = \begin{pmatrix} e^{i\chi} & 0 \\ 0 & e^{i\chi} \end{pmatrix}\begin{pmatrix} \cos(\phi/2)e^{i(\theta/2+\kappa)} & \sin(\phi/2)e^{i\theta/2} \\ \sin(\phi/2)e^{i(-\theta/2+\kappa)} & -\cos(\phi/2)e^{-i\theta/2} \end{pmatrix}\begin{pmatrix} \vec{v}_1 \\ \vec{v}_2 \end{pmatrix} \tag{234}$$

Parameter ranges?Optimally I would find a Group-theoretical justification of this representaion. χ is a global phase, which I choose to be 0, justified by the $U(1)$-gauge symmetry of quantum mechanics. This leaves three parameters. Now I can write down the projection of $H^1_{\vec{K}}$ onto $\{\vec{w}_1,\vec{w}_2\}$. I write the resulting hermitian 2x2 matrix in the Pauli-matrix basis (again there is no σ_0-part):

$$\langle H^1_{\vec{K}}\rangle_{\{\vec{w}_1,\vec{w}_2\}} = \sigma_x \left[M_{11}\sin(\phi)\cos(\theta) + M_{12}\left(\sin(\kappa)\sin(\theta) - \cos(\phi)\cos(\kappa)\cos(\theta)\right)\right] \tag{235}$$

$$+\sigma_y \left[M_{11}\sin(\phi)\sin(\theta) - M_{12}\left(\sin(\kappa)\cos(\theta) + \cos(\phi)\cos(\kappa)\sin(\theta)\right)\right] \tag{236}$$

$$+\sigma_z \left[M_{11}\cos(\phi) + M_{12}\sin(\phi)\cos(\kappa)\right] \tag{237}$$

One might complain that the equations above dont reference the Fermi velocity. However, as long as we only make statements using ratios of M-components we still obtain the desired result. Now, (ϕ, θ, κ) can be choosen to i.e. give a projected Hamiltonian of the form

$$\langle H^1_{\vec{K}}\rangle_{\{\vec{w}_1,\vec{w}_2\}} \stackrel{!}{=} v_F\big(p_x\sigma_x + p_y\sigma_y\big). \tag{238}$$

A partial solution is

$$\begin{aligned} \cos(\phi) &= 0 \\ \cos(\kappa) &= 0 \\ \tan(\theta) &= -\frac{M^x_{12}}{M^y_{12}}. \end{aligned} \tag{239}$$

The above is not sufficient to completely fix ϕ and κ as both $\pi/2$ and $-\pi/2$ are still correct solutions, yet only one combination gives the correct result. However, the partial solutions ensures that the σ_z component vanishes in every case. At the moment I just have to try all 4 possibilities until I found a systematic solution. I expect that the answer lies in the signs of the M_{ij}.

One could (for example) also pick the form

$$\langle H^1_{\vec{K}}\rangle_{\{\vec{w}_1,\vec{w}_2\}} \stackrel{!}{=} v_F\big(p_x\sigma_x + p_y\sigma_z\big). \tag{240}$$

A partial solution for this is

$$\begin{aligned} \tan(\phi) &= \sqrt{-\frac{M^x_{11}M^y_{12}}{M^y_{11}M^x_{12}}} \\ \sin(\kappa) &= 0 \\ \sin(\theta) &= 0. \end{aligned} \tag{241}$$

Again similar problem as above, κ and θ are not completely fixed yet. In what follows I will usually apply the second form as a consistency check.

5.1.2 Straining the Tight-Binding Model

I will only give the relevant equations since I have used this method prior. Straining the tight binding model works as follows:

$$t(\vec{\delta}_{ij})e^{i\vec{K}\cdot\vec{\delta}_{ij}} \approx te^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(1 + i\vec{p}\cdot\vec{\delta}_{ij} - \beta\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) + i\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)\right) \tag{242}$$

and

$$\begin{aligned} t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} + t(-\vec{\delta}_{ij})e^{-i\vec{K}\cdot\vec{\delta}_{ij}} \approx& 2t\cos\left(\vec{q}\cdot\vec{\delta}_{ij}\right) - 2t\sin\left(\vec{q}\cdot\vec{\delta}_{ij}\right)\left(\vec{p}\cdot\vec{\delta}_{ij}\right) \\ &-2\beta t\cos\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \\ &-2t\sin\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \end{aligned} \tag{243}$$

and

$$\begin{aligned} t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} - t(-\vec{\delta}_{ij})e^{-i\vec{q}\cdot\vec{\delta}_{ij}} \approx & 2it\sin\left(\vec{K}\cdot\vec{\delta}_{ij}\right) + 2it\cos\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{p}\cdot\vec{\delta}_{ij}\right) \\ & -2i\beta t\sin\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \\ & +2it\cos\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \end{aligned} \tag{244}$$

I will usually split everything into two parts:

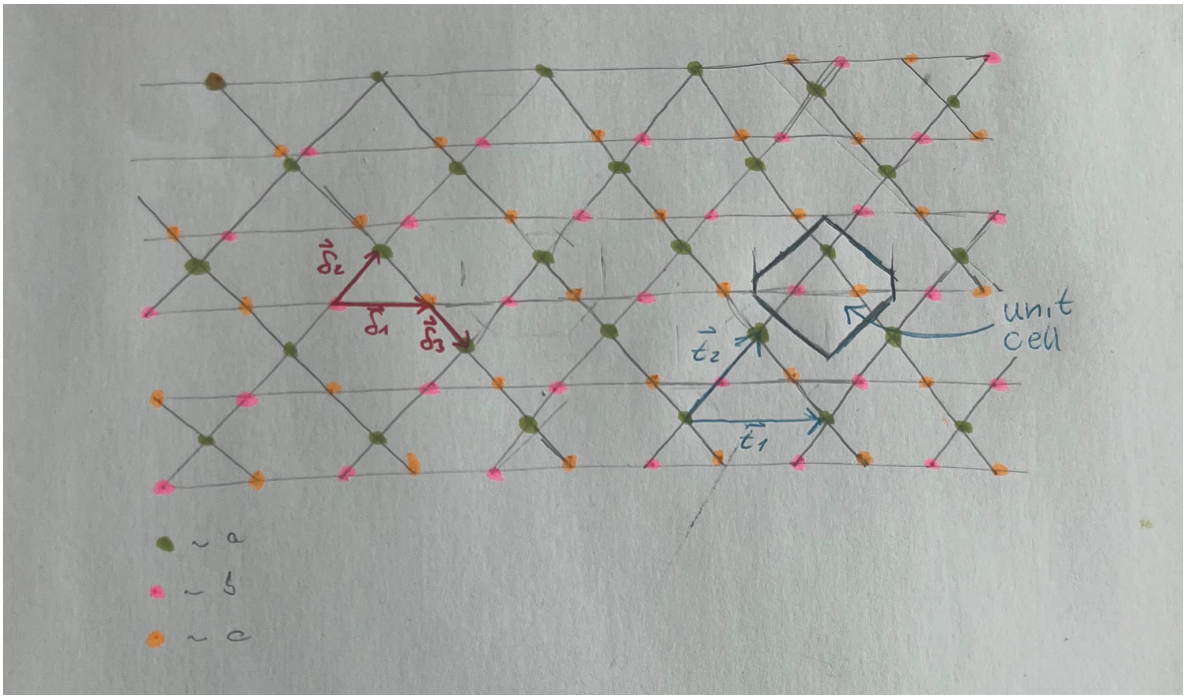
$$\delta H_0 \quad \text{and} \quad \delta H_1, \tag{245}$$

where δH_i is of order i in β.

5.2 0-flux State

5.2.1 Real Space

The bare Kagome lattice is a triangular Bravais lattice with a three site unit cell (called a, b and c-sites below). The Wigner Seitz-cell is a hexagon (with full symmetries, even if the drawing suggests otherwise :)). Geometrically speaking the lattice consists of corner-sharing equilateral triangles.



In all that follows I normalize all distances wrt the distance between nearest neighbours. The nearest neighbour vectors are (see cartoon)

$$\vec{\delta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{\delta}_2 = \frac{1}{2}\begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad \vec{\delta}_3 = \frac{1}{2}\begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix} \tag{246}$$

and the primitive translations

$$\vec{t}_1 = 2\vec{\delta}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \vec{t}_2 = 2\vec{\delta}_2 = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}. \tag{247}$$

The tight binding hamiltonain is

$$H = t\sum_{i\sigma}\left(a^\dagger_{i\sigma}c_{i\sigma} + c^\dagger_{i\sigma}b_{i\sigma} + b^\dagger_{i\sigma}a_{i\sigma} + c^\dagger_{i\sigma}b_{i+t_1;\sigma} + c^\dagger_{i\sigma}a_{i+t_1-t_2;\sigma} + a^\dagger_{i\sigma}b_{i+t_2;\sigma} + \text{h.c.}\right) \tag{248}$$

The first index refers to the unit cell and the second one to the "species" of the particle, the latter being a product of the parton decomposition.

5.2.2 Reciprocal Space and Bands

The recipricoal lattice vectors are

$$\vec{b}_1 = \pi\begin{pmatrix}1\\-1/\sqrt{3}\end{pmatrix}, \quad \vec{b}_2 = \pi\begin{pmatrix}0\\2/\sqrt{3}\end{pmatrix} \tag{249}$$

The first Brillouin zone is also a hexagon (rotated by 90° wrt to the Wigner-Seitz cell). To find H in reciprocal space I do a Fourier transform:

$$a_{i\sigma} = \frac{1}{\sqrt{N}}\sum_k e^{i\vec{k}\vec{R}_i}a_{k\sigma}, \tag{250}$$

where N is the number of unit cells. This yields

$$H = 2t\sum_{q\sigma}\begin{pmatrix}a^\dagger_{q\sigma} & b^\dagger_{q\sigma} & c^\dagger_{q\sigma}\end{pmatrix}\begin{pmatrix}0 & \cos\left(\vec{q}\cdot\vec{\delta}_2\right) & \cos\left(\vec{q}\cdot\vec{\delta}_3\right)\\ \cos\left(\vec{q}\cdot\vec{\delta}_2\right) & 0 & \cos\left(\vec{q}\cdot\vec{\delta}_1\right)\\ \cos\left(\vec{q}\cdot\vec{\delta}_3\right) & \cos\left(\vec{q}\cdot\vec{\delta}_1\right) & 0\end{pmatrix}\begin{pmatrix}a_{q\sigma}\\b_{q\sigma}\\c_{q\sigma}\end{pmatrix}. \tag{251}$$

Bands

The three bands are

$$\varepsilon_1(\vec{q}) = t\left(1 - \sqrt{4\cos^2(q_x) + 4\cos(q_x)\cos\left(\sqrt{3}q_y\right) + 1}\right) \tag{252}$$

$$\varepsilon_2(\vec{q}) = t\left(1 + \sqrt{4\cos^2(q_x) + 4\cos(q_x)\cos\left(\sqrt{3}q_y\right) + 1}\right) \tag{253}$$

$$\varepsilon_3(\vec{q}) = -2t, \tag{254}$$

see [**?**]. Interestingly, the top band is completely flat.

Dirac points

The other two bands touch when

$$\cos^2(q_x) + \cos(q_x)\cos\left(\sqrt{3}q_y\right) + 1/4 = 0 \tag{255}$$

$$\Rightarrow \cos(q_x) = \frac{1}{2}\left(-\cos\left(\sqrt{3}q_y\right) \pm \sqrt{\cos^2(\sqrt{3}q_y) - 1}\right) \tag{256}$$

A handy solution to this equation is

$$\begin{aligned} q_x &= (-1)^s\frac{2\pi}{3} + m_1\pi\\ q_y &= \frac{\pi}{\sqrt{3}}\left(2m_2 - m_1\right), \end{aligned} \tag{257}$$

with $s \in 0, 1$ and $m_1, m_2 \in \mathbb{N}$. These touching points lie at the corners of the Brillouin zone. They are all symmetry-equivalent. Representives for the time-revearsal partners in the first Brillouin zone are

$$\vec{K} = \begin{pmatrix} 2\pi/3 \\ 0 \end{pmatrix}, \quad s = 0,\ m_1 = 0,\ m_2 = 0 \tag{258}$$

$$\vec{K}' = \begin{pmatrix} -2\pi/3 \\ 0 \end{pmatrix}, \quad s = 1,\ m_1 = 0,\ m_2 = 0. \tag{259}$$

Small Momentum Expansion

Linearize the Hamiltonian according to

$$H(\vec{K} + \vec{p}) \approx \underbrace{H(\vec{K})}_{\equiv H^0_{\vec{K}}} + \underbrace{\vec{p} \cdot \left(\vec{\nabla}_{\vec{q}} H(\vec{q}) \right) \Big|_{\vec{q}=\vec{K}}}_{\equiv H^1_{\vec{K}}} + \mathcal{O}(p^2). \tag{260}$$

and find

$$H^0_{\vec{K}} = 2t \begin{pmatrix} 0 & \cos(K_2) & \cos(K_3) \\ \cos(K_2) & 0 & \cos(K_1) \\ \cos(K_3) & \cos(K_1) & 0 \end{pmatrix} \tag{261}$$

and

$$H^1_{\vec{K}} = -2t \begin{pmatrix} 0 & \sin(K_2)p_2 & \sin(K_3)p_3 \\ \sin(K_2)p_2 & 0 & \sin(K_1)p_1 \\ \sin(K_3)p_3 & \sin(K_1)p_1 & 0 \end{pmatrix}. \tag{262}$$

Projection

The eigenvector of $H^0_{\vec{K}}$ corresponding to the flat band reads:

$$\vec{v}_3(\vec{K}) = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}. \tag{263}$$

A possible (orthonormal) basis for the subspace orthogonal to $\vec{v}_3$ is

$$\vec{v}_1(\vec{K}) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{v}_2(\vec{K}) = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}. \tag{264}$$

Projeting H onto this basis yields the coefficients

$$\begin{aligned} M^x_{11} &= -\frac{\sqrt{3}}{2}t \\ M^y_{11} &= -\frac{3}{2}t \\ M^x_{12} &= -\frac{3}{2}t \\ M^y_{12} &= \frac{\sqrt{3}}{2}t. \end{aligned} \tag{265}$$

$$\Rightarrow v_F(\vec{K}) = \sqrt{3}t \tag{266}$$

The corresponding parameters for an effective theory of the form $H = \varepsilon_0\sigma_0 + p_x\sigma_x p_y\sigma_y$ are

$$\begin{aligned}\phi &= -\frac{\pi}{2}\\ \kappa &= -\frac{\pi}{2}\\ \theta &= \arctan\Big(\sqrt{3}\Big) = \frac{\pi}{3}.\end{aligned} \tag{267}$$

5.2.3 Strain and Pseudo-Potentials

Introducing strain yields the following additional expansion terms of H:

$$\delta H_1 = -2t\beta \begin{pmatrix} 0 & (u_{xx} + 2\sqrt{3}u_{yx} + 3u_{yy})/8 & (u_{xx} - 2\sqrt{3}u_{yx} + 3u_{yy}/8 \\ (u_{xx} + 2\sqrt{3}u_{yx} + 3u_{yy})/8 & 0 & -u_{xx}/2 \\ (u_{xx} - 2\sqrt{3}u_{yx} + 3u_{yy}/8 & -u_{xx}/2 & 0 \end{pmatrix} \tag{268}$$

$$\delta H_2 = -2t \begin{pmatrix} 0 & \pi(\frac{u_{xx}}{\sqrt{3}} + u_{yx})/2 & \pi(\frac{u_{xx}}{\sqrt{3}} - u_{yx})/2 \\ \pi(\frac{u_{xx}}{\sqrt{3}} + u_{yx})/2 & 0 & \pi\frac{u_{xx}}{\sqrt{3}} \\ \pi(\frac{u_{xx}}{\sqrt{3}} - u_{yx})/2 & \pi\frac{u_{xx}}{\sqrt{3}} & 0 \end{pmatrix}. \tag{269}$$

Projecting these onto the correct basis gives

$$\langle \delta H_1 \rangle_{\{\vec{w}_1, \vec{w}_2\}} = -\frac{\beta}{2}(u_{xx} + u_{yy})\sigma_0 + \sqrt{3}\frac{\beta}{2\sqrt{3}}(u_{yy} - u_{xx})\sigma_x + \sqrt{3}\frac{\beta}{\sqrt{3}}u_{yx}\sigma_y \tag{270}$$

$$\langle \delta H_0 \rangle_{\{\vec{w}_1, \vec{w}_2\}} = \sqrt{3}\frac{2\pi}{3}u_{xx}\sigma_x + \sqrt{3}\frac{2\pi}{3}u_{yx}\sigma_y. \tag{271}$$

Thus, the final result is

$$\phi(\vec{K}) = -\frac{\beta}{2}(u_{xx} + u_{yy}) \tag{272}$$

$$\vec{A}(\vec{K}) = \frac{\beta}{2\sqrt{3}} \begin{pmatrix} u_{yy} - u_{xx} \\ 2u_{yx} \end{pmatrix} \tag{273}$$

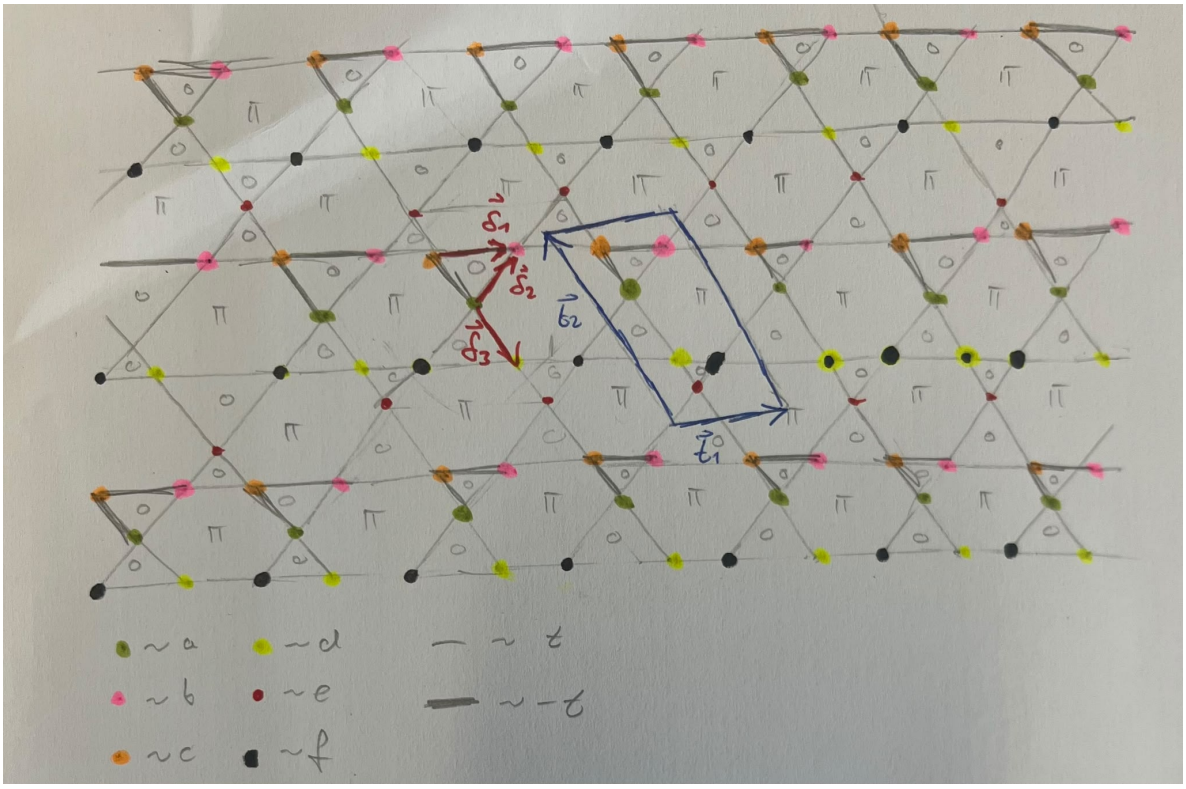
This is the result they found in [**?**].

$$\Rightarrow \vec{B}(\vec{K}) = -\frac{\beta}{2\sqrt{3}}(\partial_x\partial_y U_x + [2\partial_x^2 - \partial_y^2]U_y) \tag{274}$$

5.3 π-flux State

5.3.1 Real Space

This system is a rectengular Bravais-lattice with a six site unit cell. It emerges from the bare Kagome lattice through unit cell doubling. Even though the Wigner-Seitz-cell is a rectangle I will work with the unit cell shown below.



The nearest neighbour vectors remain unchanged

$$\vec{\delta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{\delta}_2 = \frac{1}{2}\begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad \vec{\delta}_3 = \frac{1}{2}\begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix}. \tag{275}$$

The new primitve translations are

$$\vec{t}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \vec{t}_2 = \begin{pmatrix} -2 \\ 2\sqrt{3} \end{pmatrix} \tag{276}$$

The real space (effective) tight-binding hamiltonian is

$$\begin{aligned} H = t \sum_{i\sigma} \Big(& - a^\dagger_{i\sigma} c_{i\sigma} - c^\dagger_{i\sigma} b_{i\sigma} + b^\dagger_{i\sigma} a_{i\sigma} + a^\dagger_{i\sigma} d_{i\sigma} + d^\dagger_{i\sigma} e_{i\sigma} + e^\dagger_{i\sigma} f_{i\sigma} + f^\dagger_{i\sigma} d_{i\sigma} \\ & + b^\dagger_{i\sigma} c_{i+t_1;\sigma} + b^\dagger_{i\sigma} e_{i+t_1+t_2;\sigma} + f^\dagger_{i\sigma} d_{i+t_1;\sigma} + f^\dagger_{i\sigma} a_{i+t_1;\sigma} + e^\dagger_{i\sigma} c_{i-t_2;\sigma} + \text{h.c.} \Big) \end{aligned} \tag{277}$$

5.3.2 Reciprocal Space and Bands

The reciprocal lattice vectrors are

$$\vec{b}_1 = \pi \begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix} \quad \vec{b}_2 = \pi \begin{pmatrix} 0 \\ 1/\sqrt{3} \end{pmatrix} \tag{278}$$

I do a Fouriertransform

$$\Psi_{q\sigma} \equiv \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \\ c_{q\sigma} \\ e_{q\sigma} \\ f_{q\sigma} \\ d_{q\sigma} \end{pmatrix} = \frac{1}{\sqrt{N}} \sum_i e^{i\vec{q}\vec{R}_i} \begin{pmatrix} a_{i\sigma} \\ b_{i\sigma} \\ c_{i\sigma} \\ e_{i\sigma} \\ f_{i\sigma} \\ d_{i\sigma} \end{pmatrix} \tag{279}$$

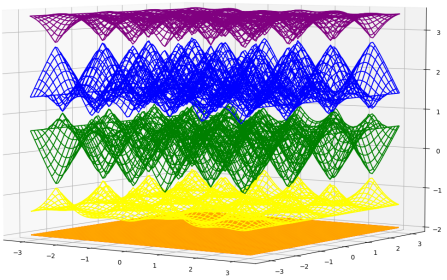
(N is the numer of unit cells.) The definition of Ψ is a bit awkward. If I write this down somewhere serious I should just label the sites differently.

$$H = t \sum_{q,\sigma} \Psi^\dagger_{q\sigma} \begin{pmatrix} 0 & e^{iq_2} & -e^{-iq_3} & 0 & e^{-iq_2} & e^{iq_3} \\ e^{-iq_2} & 0 & 2i\sin(q_1) & e^{iq_2} & 0 & 0 \\ -e^{iq_3} & -2i\sin(q_1) & 0 & e^{-iq_3} & 0 & 0 \\ 0 & e^{-iq_2} & e^{iq_3} & 0 & e^{iq_2} & e^{-iq_3} \\ e^{iq_2} & 0 & 0 & e^{-iq_2} & 0 & 2\cos(q_1) \\ e^{-iq_3} & 0 & 0 & e^{iq_3} & 2\cos(q_1) & 0 \end{pmatrix} \Psi_{q\sigma}, \tag{280}$$

where the shorthand $q_i \equiv \vec{q} \cdot \vec{\delta}_i$ was used.

Bands

The dispersion relation is shown in the following picture



This plot was generatey by numerically calculating the eigenvalues of H on a (q_x, q_y)-grid. The lowest band (flat band) is doubly degenerate.

The characteristic polynomial of 280 is (this was generate by mathematica)

$$x^6 - 12x^4 - 8x^3 + x^2(30 + 2\Delta_{\vec{q}}) + x(24 + 8\Delta_{\vec{q}}) + 8\Delta_{\vec{q}} - 8 \tag{281}$$

units! with

$$\Delta_{\vec{q}} = -\cos(4q_2) + \cos(4q_3) - \cos(4q_1). \tag{282}$$

I should verify this uisng Cayley-Hamilton. To derive analytical expressions for the bands I first divide out the two flat bands (i.e. polynomial division by $(x+2)^2$). The result is

$$x^4 - 4x^3 + 8x - 2 + 2\Delta_{\vec{q}}. \tag{283}$$

To make progress I postulate (judging from the numeriacl plot of the dispersion) that the above polynomial can be written as a product of two quadratic polynomials. Furtheremore, the solutions of said qaudratic equations must be centered around t. The p-q forumla suggests that the linear coefficents of both quadratic polynomials must therefore be $-2t$.

$$\begin{aligned}\Rightarrow x^4 - 4x^3 + 8x - 2 + 2\Delta_{\vec{q}} &= (x^2 - 2x + a) \cdot (x^2 - 2x + b) \\ &= x^4 - 4x^3 + x^2(4 + a + b) - 2x(a + b) + ab.\end{aligned} \tag{284}$$

$$\Rightarrow a + b \stackrel{!}{=} -4, \quad ab \stackrel{!}{=} -2 + 2\Delta_{\vec{q}}. \tag{285}$$

These equations are solved by

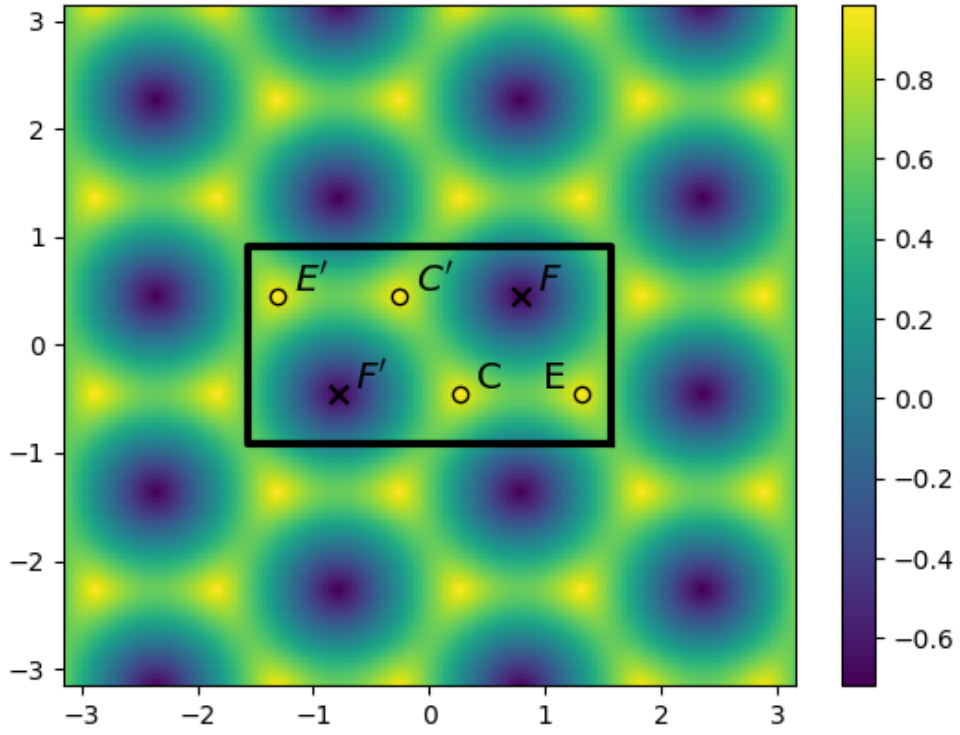
$$\begin{aligned}a_{1,2} &= -2 \pm \sqrt{6 - 2\Delta_{\vec{q}}} \\ b_{1,2} &= -2 \mp \sqrt{6 - 2\Delta_{\vec{q}}}.\end{aligned} \tag{286}$$

Both solution lead to the same result (a and b are exchanged in the second soltuion wrt the first one). This gives the following expressions for the bands

$$\begin{aligned}\varepsilon_1(\vec{q}) &= -2t \\ \varepsilon_2(\vec{q}) &= -2t \\ \varepsilon_3(\vec{q}) &= \left(1 - \sqrt{3 + \sqrt{6 - 2\Delta_{\vec{q}}}}\right) t \\ \varepsilon_4(\vec{q}) &= \left(1 - \sqrt{3 - \sqrt{6 - 2\Delta_{\vec{q}}}}\right) t \\ \varepsilon_5(\vec{q}) &= \left(1 + \sqrt{3 - \sqrt{6 - 2\Delta_{\vec{q}}}}\right) t \\ \varepsilon_6(\vec{q}) &= \left(1 + \sqrt{3 + \sqrt{6 - 2\Delta_{\vec{q}}}}\right) t.\end{aligned} \tag{287}$$

$$\Delta_{\vec{q}} = -\cos(4q_2) + \cos(4q_3) - \cos(4q_1). \tag{288}$$

Dirac Points



Shown above is the 4th band (green band). The black rectangle is the First Brillouin zone. Also shown are the position of the various band-touching points. F the Dirac points at 1/2- (and 5/6-) filling. At 2/3-filling the Dirac points are at C and E.

Now I derive the coordiantes. It is clear from the expressions above that bands 3 and 4 as well as bands 5 and 6 touch iff

$$\begin{aligned} & 6 - 2\Delta_{\vec{q}} = 0 \\ \Leftrightarrow & -\cos(4q_2) + \cos(4q_3) - \cos(4q_1) = 3 \\ \Leftrightarrow & \, 2\sin(2q_x)\sin\Big(2\sqrt{3}q_y\Big) - \cos(4q_x) = 3. \end{aligned} \tag{289}$$

This can easily be solved considering the ranges of sine and cosine. A possible parametrization of the solution is

$$q_x = (-1)^s \frac{\pi}{4} + m_1\pi \tag{290}$$

$$q_y = (-1)^s \frac{\pi}{4\sqrt{3}} + (m_1 + m_2)\frac{\pi}{\sqrt{3}}, \tag{291}$$

where

$$s \in [0, 1], \quad m_1, m_2 \in \mathbb{Z}.$$

Both $(s = 0, m_1 = 0, m_2 = 0)$ and $(s = 1, m_1 = 0, m_2 = 0)$ are in the first Brillouin zone:

$$\vec{F} = \frac{\pi}{4}\begin{pmatrix}1\\1/\sqrt{3}\end{pmatrix}, \quad \vec{F}' = -\frac{\pi}{4}\begin{pmatrix}1\\1/\sqrt{3}\end{pmatrix} \tag{292}$$

.
On the other hand, bands 4 and 5 touch when

$$\begin{aligned}\sqrt{3-\sqrt{6-2\Delta_{\vec{q}}}} &= 0\\ \Leftrightarrow 2\Delta_{\vec{q}} &= -3\\ \Leftrightarrow 4\sin(2q_x)\sin\Big(2\sqrt{3}q_y\Big) - 2\cos(4q_x) &= -3\end{aligned} \tag{293}$$

Using $\cos(2x) = 1 - 2\sin^2(x)$ I obtain

$$\begin{aligned}\sin^2(2q_x) + \sin(2q_x)\sin\Big(2\sqrt{3}q_y\Big) + \frac{1}{4} &= 0\\ \Rightarrow \sin(2q_x) = \frac{1}{2}\left(-\sin\Big(2\sqrt{3}q_y\Big) \pm \sqrt{\sin^2(2\sqrt{3}q_y) - 1}\right)\end{aligned} \tag{294}$$

The expression under the square root gives a condition for q_y. It tourns out there are two distinct sets of solutions (meaning there are two distinct sets of symmetry-equivalent touching points).

$$q_x = (-1)^s\frac{\pi}{12} + m_1\pi \tag{295}$$

$$q_y = (-1)^{s+1}\frac{\pi}{4\sqrt{3}} + (m_1+m_2)\frac{\pi}{\sqrt{3}} \tag{296}$$

and

$$q_x = (-1)^s\frac{5\pi}{12} + m_1\pi \tag{297}$$

$$q_y = (-1)^{s+1}\frac{\pi}{4\sqrt{3}} + (m_1+m_2)\frac{\pi}{\sqrt{3}}, \tag{298}$$

with the usual parmeters. Once more, there are two of each in the first Brillouin zone corresponding to $(s=0, m_1=0, m_2=0)$ and $(s=1, m_1=0, m_2=0)$:

$$\begin{aligned}\vec{C} &= \frac{\pi}{12}\begin{pmatrix}1\\-\sqrt{3}\end{pmatrix}, \quad \vec{C}' = \frac{\pi}{12}\begin{pmatrix}-1\\\sqrt{3}\end{pmatrix},\\ \vec{E} &= \frac{\pi}{12}\begin{pmatrix}5\\-\sqrt{3}\end{pmatrix}, \quad \vec{E}' = \frac{\pi}{12}\begin{pmatrix}-5\\\sqrt{3}\end{pmatrix}.\end{aligned} \tag{299}$$

Linearization of H/ Field Theory

To derive an effective (linear) theory in the vicinity of the touching points I linearize the Hamiltonian in $\vec{q}$.

$$H(\vec{K}+\vec{p}) \approx \underbrace{H(\vec{K})}_{\equiv H^0_{\vec{K}}} + \underbrace{\vec{p}\cdot\left(\vec{\nabla}_{\vec{q}}H(\vec{q})\right)\Big|_{\vec{q}=\vec{K}}}_{\equiv H^1_{\vec{K}}} + \mathcal{O}(p^2). \tag{300}$$

$$H^0_{\vec{K}} = \begin{pmatrix} 0 & e^{iK_2} & -e^{-iK_3} & 0 & e^{-iK_2} & e^{iK_3} \\ e^{-iK_2} & 0 & 2i\sin(K_1) & e^{iK_2} & 0 & 0 \\ -e^{iK_3} & -2i\sin(K_1) & 0 & e^{-iK_3} & 0 & 0 \\ 0 & e^{-iK_2} & e^{iK_3} & 0 & e^{iK_2} & e^{-iK_3} \\ e^{iK_2} & 0 & 0 & e^{-iK_2} & 0 & 2\cos(K_1) \\ e^{-iK_3} & 0 & 0 & e^{iK_3} & 2\cos(K_1) & 0 \end{pmatrix} \tag{301}$$

$$H^1_{\vec{K}} = \begin{pmatrix} 0 & ie^{iK_2}p_2 & ie^{-iK_3}p_3 & 0 & -ie^{-iK_2}p_2 & ie^{iK_3}p_3 \\ -ie^{-iK_2}p_2 & 0 & 2i\cos(p_1) & ie^{iK_2}p_2 & 0 & 0 \\ -ie^{iK_3}p_3 & -2i\cos(p_1) & 0 & -ie^{-iK_3}p_3 & 0 & 0 \\ 0 & -ie^{-iK_2}p_2 & ie^{iK_3}p_3 & 0 & ie^{iK_2}p_2 & -ie^{-iK_3}p_3 \\ ie^{iK_2}p_2 & 0 & 0 & -ie^{-iK_2}p_2 & 0 & -2\sin(p_1) \\ -ie^{-iK_3}p_3 & 0 & 0 & ie^{iK_3}p_3 & -2\cos(p_1) & 0 \end{pmatrix} \tag{302}$$

$$K_i \equiv \vec{K}\cdot\vec{\delta}_i$$
$$p_i \equiv \vec{p}\cdot\vec{\delta}_i$$

5.3.3 1/2-Filling

The Dirac points are at

$$\vec{F} = \frac{\pi}{4}\begin{pmatrix}1\\1/\sqrt{3}\end{pmatrix}, \quad \vec{F}' = -\frac{\pi}{4}\begin{pmatrix}1\\1/\sqrt{3}\end{pmatrix}. \tag{303}$$

I focus on $\vec{F}$ for now. At $\vec{F}$ I choose the basis

$$\vec{v}_3(\vec{F}) = \frac{1}{2\sqrt{6-3\sqrt{3}}}\left(\tfrac{1-\sqrt{3}}{2}+\tfrac{3-\sqrt{3}}{2}i, \quad \sqrt{2}(1-\sqrt{3}), \quad (-2+\sqrt{3})i, \quad \tfrac{1-\sqrt{3}}{2}+\tfrac{\sqrt{3}-3}{2}i, \quad 0, \quad 1\right)^T \tag{304}$$

$$\vec{v}_4(\vec{F}) = \frac{1}{2\sqrt{6(7-4\sqrt{3})}}\left((3-2\sqrt{3})+(-2+\sqrt{3})i, \quad 0, \quad i(1-\sqrt{3}), \quad 3-2\sqrt{3}+(2-\sqrt{3})i, \quad 2\sqrt{2}(2-\sqrt{3}), \quad 5-3\sqrt{3}\right)^T \tag{305}$$

I obtain

$$\begin{aligned} M^x_{33} &= \frac{1-\sqrt{3}}{2}t \\ M^y_{33} &= -\frac{1+\sqrt{3}}{2}t \\ M^x_{34} &= -\frac{1+\sqrt{3}}{2}t \\ M^y_{34} &= -\frac{1-\sqrt{3}}{2}t \end{aligned} \tag{306}$$

$$\Rightarrow v_F^{\frac{1}{2}}(\vec{F}) = \sqrt{2}t \tag{307}$$

The correct parameters for a theory of the form $H = \epsilon_0\sigma_0 + p_x\sigma_x + p_y\sigma_y$ are

$$\begin{aligned} \phi &= -\frac{\pi}{2} \\ \kappa &= -\frac{\pi}{2} \\ \theta &= \arctan\left(2+\sqrt{3}\right) = \frac{5\pi}{12} \end{aligned} \tag{308}$$

The projections onto these "correct" basis vectors are

$$\begin{aligned}\langle \delta H_1 \rangle_{\vec{w}_1,\vec{w}_2} &= -\frac{\beta}{2}(1-\sqrt{3})(u_{xx}+u_{yy})\sigma_0 \\ \langle \delta H_0 \rangle_{\vec{w}_1,\vec{w}_2} &= \sqrt{2}\frac{\pi}{4}\left(u_{xx}+\frac{u_{yx}}{\sqrt{3}}\right)\sigma_x + \sqrt{2}\frac{\pi}{4}\left(u_{xy}+\frac{u_{yy}}{\sqrt{3}}\right)\sigma_y.\end{aligned} \tag{309}$$

Hence, there is no pseudomagnetic-field but a scalar potential:

$$\phi^{\frac{1}{2}}(\vec{F}) = -\frac{\beta}{2}(1-\sqrt{3})(u_{xx}+u_{yy})\sigma_0 \tag{310}$$

$$\vec{A}^{\frac{1}{2}}(\vec{F}) = 0. \tag{311}$$

5.3.4 2/3-Filling

The Dirac points are at

$$\begin{aligned}\vec{C} &= \frac{\pi}{12}\begin{pmatrix}1\\-\sqrt{3}\end{pmatrix}, \quad \vec{C}' = \frac{\pi}{12}\begin{pmatrix}-1\\\sqrt{3}\end{pmatrix}, \\ \vec{E} &= \frac{\pi}{12}\begin{pmatrix}5\\-\sqrt{3}\end{pmatrix}, \quad \vec{E}' = \frac{\pi}{12}\begin{pmatrix}-5\\\sqrt{3}\end{pmatrix}.\end{aligned} \tag{312}$$

I will only consider $\vec{E}$ and $\vec{C}$.

$\tilde{\mathbf{C}}$

At $\vec{C}$ I choose the vectors

$$\vec{v}_4(\vec{C}) = \frac{1}{\sqrt{6}\sqrt{5\sqrt{3}+9}}\left(\frac{1+\sqrt{3}}{\sqrt{2}} - i\sqrt{2}(2+\sqrt{3}) \quad 2+\sqrt{3} \quad i\sqrt{2}(2+\sqrt{3}) \quad \frac{1+\sqrt{3}}{\sqrt{2}} + i\sqrt{2}(2+\sqrt{3}) \quad 1 \quad 0\right) \tag{313}$$

$$\vec{v}_5(\vec{C}) = \frac{\sqrt{2}}{\sqrt{3\sqrt{3}(1+\sqrt{3})}}\left(-\frac{1}{2} \quad -\frac{3}{2\sqrt{2}}(1+\sqrt{3}) \quad i\frac{1+\sqrt{3}}{2} \quad -\frac{1}{2} \quad \frac{1+\sqrt{3}}{2\sqrt{2}} \quad 1\right) \tag{314}$$

.
The matrix elements are

$$\begin{aligned}M^x_{44} &= -\frac{1+\sqrt{3}}{2}t \\ M^y_{44} &= -\frac{1-\sqrt{3}}{2}t \\ M^x_{45} &= -\frac{1-\sqrt{3}}{2}t \\ M^y_{45} &= \frac{1+\sqrt{3}}{2}t\end{aligned} \tag{315}$$

$$\Rightarrow v_F^{\frac{2}{3}}(\vec{C}) = \sqrt{2}t \tag{316}$$

The correct parameters for a theory of the form $\varepsilon_0\sigma_0 + p_x\sigma_x + p_y\sigma_y$ are

$$\begin{aligned}\phi &= -\frac{\pi}{2}\\ \kappa &= -\frac{\pi}{2}\\ \theta &= -\arctan\Big(2-\sqrt{3}\Big) = -\frac{\pi}{12}.\end{aligned} \tag{317}$$

The resulting projections are

$$\langle \delta H_1\rangle_{\vec{w}_4,\vec{w}_5} = -\frac{\beta}{2}(u_{xx}+u_{yy})\sigma_0 + \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{yy}-u_{xx})\sigma_x + \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{xy}+u_{yx})\sigma_y \tag{318}$$

$$\langle \delta H_0\rangle_{\vec{w}_4,\vec{w}_5} = \sqrt{2}\frac{\pi}{12}(u_{xx}-\sqrt{3}u_{yx})\sigma_x + \sqrt{2}\frac{\pi}{12}(u_{xy}-\sqrt{3}u_{yy})\sigma_y \tag{319}$$

For verification pourposes I will also consider a second theory of the form $\varepsilon_0\sigma_0 + p_x\sigma_x + p_y\sigma_z$, the corresponding parameters are

$$\begin{aligned}\phi &= \arctan\Big(7+4\sqrt{3}\Big) = \frac{5\pi}{12}\\ \kappa &= 0\\ \theta &= \pi.\end{aligned} \tag{320}$$

In this case I find the projections

$$\langle \delta H_1\rangle_{\vec{w}_4,\vec{w}_5} = -\frac{\beta}{2}(u_{yy}+u_{xx})\sigma_0 + \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{yy}-u_{xx})\sigma_x + \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{xy}+u_{yx})\sigma_z \tag{321}$$

$$\langle \delta H_0\rangle_{\vec{w}_4,\vec{w}_5} = ? \tag{322}$$

The two cases agree on

$$\phi^{\frac{2}{3}}(\vec{C}) = -\frac{\beta}{2}(u_{xx}+u_{yy}) \tag{323}$$

$$\vec{A}^{\frac{2}{3}}(\vec{C}) = \frac{\beta}{2\sqrt{3}}\begin{pmatrix}u_{yy}-u_{xx}\\ u_{xy}+u_{yx}\end{pmatrix}. \tag{324}$$

$$\Rightarrow B^{\frac{2}{3}}(\vec{C}) = \frac{\beta}{2\sqrt{3}}\left(2\partial_x\partial_y U_x + \left[\partial_x^2-\partial_y^2\right]U_y\right) \tag{325}$$

.

$\tilde{\mathbf{E}}$

At $\vec{E}$ I choose the basis vectors

$$\vec{v}_4(\vec{E}) = \frac{1}{\sqrt{6(9-5\sqrt{3})}}\begin{pmatrix}\sqrt{2}(2-\sqrt{3}) + i\frac{1-\sqrt{3}}{\sqrt{2}} & -2+\sqrt{3} & i\sqrt{2}(2-\sqrt{3}) & \sqrt{2}(2-\sqrt{3}) - i\frac{1-\sqrt{3}}{\sqrt{2}} & 1 & 0\end{pmatrix} \tag{326}$$

$$\vec{v}_5(\vec{E}) = \frac{\sqrt{2}}{\sqrt{3(3-\sqrt{3})}}\begin{pmatrix}\frac{i}{2} & \frac{\sqrt{3}}{2\sqrt{2}}(1-\sqrt{3}) & -i\frac{1-\sqrt{3}}{2} & -\frac{i}{2} & -\frac{1-\sqrt{3}}{2\sqrt{2}} & 1\end{pmatrix}. \tag{327}$$

The projection coefficents are

$$
\begin{aligned}
M^x_{44} &= -\frac{1-\sqrt{3}}{2}t \\
M^y_{44} &= \frac{1+\sqrt{3}}{2}t \\
M^x_{45} &= -\frac{1+\sqrt{3}}{2}t \\
M^y_{45} &= -\frac{1-\sqrt{3}}{2}t.
\end{aligned} \tag{328}
$$

$$
\Rightarrow v_F^{\frac{2}{3}}(\vec{E}) = \sqrt{2}t \tag{329}
$$

The correct parameters are

$$
\begin{aligned}
\phi &= \pi/2 \\
\kappa &= -\pi/2 \\
\theta &= \arctan\left(2+\sqrt{3}\right) = \frac{5\pi}{12}.
\end{aligned} \tag{330}
$$

The cooresponding projections are

$$
\langle \delta H_1 \rangle_{\vec{w}_4,\vec{w}_5} = -\frac{\beta}{2}(u_{xx}+u_{yy})\sigma_0 + \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{xx}-u_{yy})\sigma_x - \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{xy}+u_{yx})\sigma_y \tag{331}
$$

$$
\langle \delta H_0 \rangle_{\vec{w}_4,\vec{w}_5} = \sqrt{2}\frac{\pi}{12}\left(5u_{xx}-\sqrt{3}u_{yx}\right)\sigma_x + \sqrt{2}\frac{\pi}{12}(5u_{xy}-\sqrt{3}u_{yy})\sigma_y. \tag{332}
$$

Therefore,

$$
\phi^{\frac{2}{3}}(\vec{E}) = -\frac{\beta}{2}(u_{xx}+u_{yy}) \tag{333}
$$

$$
\vec{A}^{\frac{2}{3}}(\vec{E}) = \frac{\beta}{2\sqrt{3}}\begin{pmatrix} u_{xx}-u_{yy} \\ -(u_{xy}+u_{yx}) \end{pmatrix} \tag{334}
$$

It turns out $\vec{A}^{\frac{2}{3}}(\vec{E}) = -\vec{A}^{\frac{2}{3}}(\vec{C})$.

I verified tthis result by doing the calculation for $\vec{E}+\vec{b}_2$ and obtaining the same results. However, the eigenvectors were not the same why?

5.3.5 5/6-filling

The Dirac points are once more at

$$
\vec{F} = \frac{\pi}{4}\begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix}, \quad \vec{F}' = -\frac{\pi}{4}\begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix}. \tag{335}
$$

I will do the calculations for $\vec{F}$. I choose the basis vecors

$$
\vec{v}_5(\vec{F}) = \frac{1}{\sqrt{\sqrt{6}+3\sqrt{2}}}\left(-\tfrac{1}{\sqrt{2}} - i\tfrac{2+\sqrt{3}}{\sqrt{2}} \quad -2-\sqrt{3} \quad i\sqrt{2}(1+\sqrt{3}) \quad -\tfrac{1}{\sqrt{2}} + i\tfrac{2+\sqrt{3}}{\sqrt{2}} \quad 1 \quad 0\right) \tag{336}
$$

$$
\vec{v}_6(\vec{F}) = \frac{1}{4\sqrt{3}}\left(1+\sqrt{3}+i(1-\sqrt{3}) \quad \sqrt{2}+\sqrt{6} \quad 0 \quad 1+\sqrt{3}-i(1-\sqrt{3}) \quad \sqrt{2}+\sqrt{6} \quad 4\right) \tag{337}
$$

The matrix elements are

$$
\begin{aligned}
M_{55}^{x} &= \frac{1+\sqrt{3}}{2}t \\
M_{55}^{y} &= -\frac{1-\sqrt{3}}{2}t \\
M_{56}^{x} &= \frac{1-\sqrt{3}}{2}t \\
M_{56}^{y} &= \frac{1+\sqrt{3}}{2}t.
\end{aligned} \tag{338}
$$

$$
\Rightarrow v_{F}^{\frac{5}{6}}(\vec{F}) = \sqrt{2}t \tag{339}
$$

Hence, the correct parameters are

$$
\begin{aligned}
\phi &= \pi/2 \\
\kappa &= -\pi/2 \\
\theta &= \arctan\left(2-\sqrt{3}\right) = \frac{\pi}{12}
\end{aligned} \tag{340}
$$

Applying strain yields the results

$$
\langle \delta H_1 \rangle_{\vec{w}_5,\vec{w}_6} = -\frac{\beta}{2}(1+\sqrt{3})(u_{xx}+u_{yy})\sigma_0 \tag{341}
$$

$$
\langle \delta H_0 \rangle_{\vec{w}_5,\vec{w}_6} = \sqrt{2}\frac{\pi}{4}\left(u_{xx}+\frac{u_{yx}}{\sqrt{3}}\right)\sigma_x + \sqrt{2}\frac{\pi}{4}\left(u_{xy}+\frac{u_{yy}}{\sqrt{3}}\right)\sigma_y \tag{342}
$$

$$
\phi^{\frac{5}{6}}(\vec{F}) = -\frac{\beta}{2}(1+\sqrt{3})(u_{xx}+u_{yy})\sigma_0 \tag{343}
$$

$$
\vec{A}^{\frac{5}{6}}(\vec{F}) = 0. \tag{344}
$$

Part III
other

6. mail 10.07

$$\mathcal{H} = \frac{J_H}{M'} \sum_{\langle ij \rangle} \vec{S}_i \cdot \vec{S}_j + \frac{\kappa}{(M')^2} \left(\vec{S}_i \cdot \vec{S}_j \right)^2 \tag{345}$$

transformations:

$$\text{Eric}: \ \vec{S}_i = \sum_{\alpha\beta} f^\dagger_{i\alpha} \frac{\vec{\sigma}}{2} f_{i\beta} \Rightarrow \vec{S}_i \cdot \vec{S}_j = \frac{1}{4} - \frac{1}{2} \sum_{\alpha\beta} f^\dagger_{i\alpha} f_{j\alpha} f^\dagger_{j\beta} f_{i\beta} \tag{346}$$

$$\text{Pedro}: \ S_i^{\beta\alpha} = f^\dagger_{i\beta} f_{i\alpha} - \frac{Q}{M} \delta_{\alpha\beta} \Rightarrow \vec{S}_i \cdot \vec{S}_j = \frac{M}{4} - \sum_{\alpha\beta} f^\dagger_{i\alpha} f_{j\alpha} f^\dagger_{j\beta} f_{i\beta} \tag{347}$$

difference of factor 2 for M = 2, which is my case. (Also, for me $M' = 1$ while Pedro used $M' = M$.) mean field Hamiltonains:

$$\text{Eric}: \mathcal{H} \approx -J \sum_{\langle ij \rangle} \Big[(1 + \kappa/2 - 4\kappa |\langle \chi_{ij} \rangle|^2 \langle \chi_{ij} \rangle \left(f^\dagger_{j\uparrow} f_{i\uparrow} + f^\dagger_{j\downarrow} f_{i\downarrow} \right) + \text{h.c.} \tag{348}$$

$$+ 2(-1 - \kappa/2 + 6\kappa |\langle \chi_{ij} \rangle|^2) |\langle \chi_{ij} \rangle|^2 \Big] \tag{349}$$

$$\text{Pedro}: \mathcal{H} \approx -J \sum_{\langle ij \rangle} \Big[2 + \boxed{2\kappa} - 16\kappa |\langle \chi_{ij} \rangle|^2) \langle \chi_{ij} \rangle \left(f^\dagger_{j\uparrow} (f_{i\uparrow} + f^\dagger_{j\downarrow} f_{i\downarrow} \right) + \text{h.c.} \tag{350}$$

$$+ 2(-2 \boxed{-2\kappa} + 24\kappa |\langle \chi_{ij} \rangle|^2) |\langle \chi_{ij} \rangle|^2 \Big] \tag{351}$$

Every term without a κ (coming from the bilinear exchange term) is off by a factor of 2. Every term with a κ (coming from the biquadratic exchange term) is off by a factor of 4. Hence, the only problem is the factor 2 in the fermionic representation. The boxed terms dont appear in Pedros notes as they go like M^{-1} and are therfore irrelevant in the large M limit.